Ambient Streaming Activity Sharing on Social networks

TECHNICAL FIELD

This disclosure relates to a system for sharing media consumption activity.

Background

Social interaction platforms enable users to share and interact with digital content in a social context. These platforms collect a wide range of data, including media consumption activity data, which reflects how users engage with digital content. This data can include metrics such as time spent watching videos, number of likes and comments, and browsing history.

Brief Description of the Several Views of the Drawings

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some non-limiting examples are illustrated in the figures of the accompanying drawings in which:

FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, according to some examples.

FIG. 2 is a diagrammatic representation of a digital interaction system that has both client-side and server-side functionality, according to some examples.

FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, according to some examples.

FIG. 4 is a diagrammatic representation of a message, according to some examples.

FIG. 5 illustrates a streaming activity data flow architecture, according to some examples.

FIG. 6 illustrates a unified media activity structure, according to some examples.

FIG. 7 illustrates a listening status and account linking interface, according to some examples.

FIG. 8 illustrates a map interface with listening status indicator, according to some examples.

FIG. 9 illustrates map interface views with user listening activity, according to some examples.

FIG. 10 illustrates an onboarding flow for streaming service linking, according to some examples.

FIG. 11 illustrates a sharing permission dialog for listening activity, according to some examples.

FIG. 12 illustrates a settings interface for streaming service management, according to some examples.

FIG. 13 illustrates a presence state determination and suppression flowchart, according to some examples.

FIG. 14 is a flowchart illustrating a routine (e.g., a method or process), according to some examples.

FIG. 15 illustrates a system including the head-wearable apparatus, according to some examples.

FIG. 16 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed to cause the machine to perform any one or more of the methodologies discussed herein, according to some examples.

FIG. 17 is a block diagram showing a software architecture within which examples may be implemented.

Detailed Description

The description that follows discusses illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth to provide an understanding of various examples of the disclosed subject matter. It will be evident, however, to those skilled in the art, that examples of the disclosed subject matter may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

Music consumption on streaming platforms is a activity that reflects personal taste, mood, and identity. However, the act of listening to music on a streaming platform occurs in isolation from a user's social connections on other platforms. When a user plays a song on a streaming service, that activity remains confined to the streaming service's own ecosystem. If the user wishes to share what they are listening to with friends on a separate social platform, the user must take affirmative steps to do so, for example, by manually copying a song link, composing a post or story, or attaching a media sticker to a message. Each of these actions requires the user to interrupt their listening experience, navigate between applications, and perform multiple input steps. This manual sharing process consumes user time and attention, introduces friction that discourages sharing, and results in the vast majority of listening activity never reaching a user's social graph. As a consequence, computing resources on the social platform that could otherwise be used to facilitate media-related interactions among users remain idle with respect to streaming activity, and the social platform lacks data signals that could inform content presentation or friend interaction patterns.

Additionally, existing approaches to sharing listening activity on social platforms do not integrate listening presence into a map-based, real-time social context where users already monitor the locations and activities of their friends. A user who opens a map interface on a social platform to view the locations of friends sees geographic positions and, in some cases, activity indicators such as travel mode or venue check-ins, but does not see what media content those friends are consuming. The absence of media consumption information from the map interface represents a gap in the social context available to the viewing user. To discover what a friend is listening to, the viewing user must navigate to a separate application or surface, if such information is available at all. This separation between geographic presence and media consumption presence requires additional navigation steps, increases the number of application context switches, and fragments the social context that the viewing user can obtain from a single interface.

Where some degree of listening activity sharing does exist, it tends to be constrained to a single surface within a single application. For example, a streaming service may display a sidebar on its desktop client showing what a user's contacts are listening to, but this information is not accessible on mobile devices and is not integrated into any map-based, profile-based, or feed-based interface on a separate social platform. Similarly, a social platform may allow a user to attach a song to a temporary story post, but this requires the user to create and publish the post, and the information disappears after a set expiration period. These approaches do not provide a persistent, ambient, or automatically updated representation of listening activity. They also do not account for the fact that different streaming services expose activity data in different formats, with different fields, and with different latency characteristics. A system that integrates with only a single streaming service, or that handles each streaming service's data in a bespoke manner, incurs duplicated engineering effort for each additional service and cannot present a uniform listening status to users regardless of which service their friends use. This per-service duplication wastes development resources and delays the availability of cross-platform listening visibility.

Additionally, when a user on a social platform discovers a song that a friend is listening to on one streaming service, and the discovering user subscribes to a different streaming service, there is no mechanism to save that song directly to the discovering user's own library on their own service. The user must instead manually search for the song by name on their own streaming service, which requires switching applications, entering search terms, and locating the correct track among potentially many results. This process wastes the user's time, increases the number of network requests and application switches performed on the device, and frequently results in the user abandoning the effort altogether. The absence of a cross-platform save operation tied to a standardized track identifier means that the discovery moment, which occurred on the social platform, fails to convert into a saved track on the user's streaming service, and the computational work performed to surface the listening activity to the discovering user yields no downstream action.

The disclosed examples improve the efficiency of using an electronic device by eliminating the need for a user to manually share media consumption activity across separate applications, reducing the number of user input actions, application switches, and network requests that would otherwise be required to communicate listening status to friends on a social interaction platform. Rather than requiring a user to copy a song link from a streaming application, switch to a social application, compose a post or message, and transmit that post, a sequence that consumes processor cycles, memory, network bandwidth, and battery resources on the device for each sharing instance, the disclosed examples automatically retrieve media consumption activity data from a third-party media streaming service, convert that data into a standardized activity record, determine a presence state, and render a listening status indicator on one or more surfaces of the social interaction platform without any affirmative action by the sharing user. The disclosed examples further reduce computational waste by normalizing heterogeneous data from multiple streaming service APIs into a unified schema, thereby allowing a single set of downstream processing operations, including presence state determination, privacy rule evaluation, and surface rendering, to operate on a common data structure regardless of the originating streaming service, rather than requiring separate processing pipelines for each partner integration.

Additionally, the disclosed examples reduce unnecessary network traffic and API consumption by ceasing retrieval of media consumption activity data when a user has been inactive on the social interaction platform beyond a defined inactivity threshold, and by ceasing retrieval when a user activates a private listening mode, thereby avoiding the expenditure of processing and network resources to obtain data that would not be displayed to any viewer.

The disclosed examples also address partner-specific branding and display constraints. Different third-party media streaming services 132 may impose different requirements regarding how their brand elements, logos, and service names are displayed within the social interaction platform 510. The system accommodates these constraints by associating partner-specific display rules with each partner identifier in the unified schema. The UI update module 516 consults these partner-specific display rules when rendering the listening status indicator on each surface. For example, a first partner may require that its logo be displayed at a specified minimum size and in a specified color when the listening status indicator appears within the tray element, while a second partner may require that its service name appear in text form rather than as a logo. The system stores these display rules in association with the partner identifier and applies them during the rendering process without modifying the underlying standardized activity record or the presence state determination logic. This separation of display constraints from data processing logic enables the social interaction platform 510 to comply with partner branding requirements while maintaining a single normalized data pipeline for all partners.

The disclosed examples further address the localization of user interface elements associated with the listening status indicator. The social interaction platform 510 may render the add song button 714, the connection prompt, the share permission dialog 1106, and other interactive elements in multiple languages based on the locale setting of the viewing user's device. For example, the add song button 714 may display "Add" in English, "Hinzufügen" in German, "Agregar" in Spanish, or corresponding text in other supported languages, and may transition to "Added," "Hinzugefügt," "Agregado," or corresponding text upon completion of the save operation. The social interaction platform 510 may store localized string resources for each interactive element and may select the appropriate string based on the locale identifier associated with the viewing user's device or account settings. The localization of interactive elements enables the listening status indicator and its associated interactive components to be rendered in a manner consistent with the viewing user's language preferences across all surfaces of the social interaction platform 510.

As an illustrative example, a first user ("User A") has linked a first streaming service account to User A's account on a social interaction platform by completing an OAuth-based authorization flow. User A opens the first streaming service application on User A's mobile device and begins playing a music track having the title "(It Goes Like) Nanana" by the artist "Peggy Gou," which has an international standard recording code (ISRC) of "NLZ542200042." A backend streaming integration service of the social interaction platform polls the first streaming service's application programming interface at a polling interval of, for example, 15 seconds, and receives a response containing media consumption activity data. The response includes partner-specific fields such as a partner track identifier "3WOhcATHxK1X," an artist field "Peggy Gou," an album field "I Hear You," a playback state field indicating "playing," and a timestamp of "2026-03-12T14:32:07Z." A partner-specific interface module associated with the first streaming service parses this response and maps the partner-specific fields to corresponding fields of a unified schema, generating a standardized activity record comprising a user identifier "userA12345," a track identifier "NLZ542200042," an artist identifier "Peggy Gou," an album identifier "I Hear You," a playback state of "playing," a timestamp of "2026-03-12T14:32:07Z," a session identifier "sess8837a," and a partner identifier "streamingservice_1." A presence state engine compares the timestamp against the current time, computes an elapsed duration of 8 seconds, determines that the elapsed duration is below a first threshold of 60 seconds and that the playback state indicates active playback, and classifies the presence state as a "live listening" state. The presence state engine then evaluates User A's sharing preference, which specifies "All Friends," and determines that a second user ("User B") is within User A's friend list and satisfies the visibility rule.

The social interaction platform 510 may support the extension of media consumption activity data beyond music tracks to other media types including podcasts, audiobooks, and video content. The unified media activity structure 604 described with reference to FIG. 6 may include a media type field that distinguishes between different categories of media content. When the normalization module 512 receives media consumption activity data from a third-party media streaming service 132 that includes a podcast episode, an audiobook chapter, or a video content item, the normalization module 512 may populate the media type field with a corresponding value such as "music," "podcast," "audiobook," or "video." The presence logic module 514 and the UI update module 516 may use the media type field to adjust the rendering of the listening status indicator. For example, when the media type field indicates "podcast," the callout element on the map interface may display the podcast name and episode title rather than a track name and artist name, and the icon rendered within the callout element may be a podcast-specific icon rather than a musical note icon. The add song button 714 may be relabeled or reconfigured to correspond to the media type, such as displaying "Add Episode" for podcast content. The internal media library of the social interaction platform 510 may maintain track entries for podcast episodes and audiobook chapters in addition to music tracks, enabling the cross-platform save operation to function across media types where the viewing user's linked streaming service supports the corresponding media type.

The social interaction platform 510 may also generate content recommendation signals based on aggregated media consumption activity data. When the social interaction platform 510 retrieves and normalizes media consumption activity data for multiple users within a social graph, the system may identify patterns in listening activity, such as tracks that are being played by multiple friends within a defined time period, or tracks that are frequently saved by viewing users after being surfaced through the listening status indicator. The social interaction platform 510 may use these signals to generate recommendations, such as surfacing a "Popular with Friends" label on a Sounds Topic Page or on a map venue label, indicating that a particular track or venue is associated with listening activity among the user's friends. The social interaction platform 510 may also use aggregated listening activity data, in a privacy-safe manner that does not expose individual users' listening activity beyond their configured sharing preferences, to inform content ranking or discovery features within the social interaction platform 510. These content recommendation signals may also serve as inputs to advertising or sponsored content systems, enabling the social interaction platform 510 to present music-related promotional content to users based on aggregate listening trends within their social graph without exposing any individual user's listening activity to advertisers.

On User B's device, the social interaction platform renders a map interface displaying a geographic area corresponding to Times Square, New York. An avatar representing User A is rendered at a position on the map interface corresponding to User A's geographic location at the intersection of 47th Street and 7th Avenue. A callout element is generated and positioned proximate to the avatar. The callout element displays the track name "(It Goes Like)..." and the artist name "Peggy Gou." Because User B is a user other than User A, the callout element renders a generic media icon, a musical note symbol, rather than a platform-specific icon identifying the first streaming service. Below the map, a label displays "Henry now," indicating User A's display name and that User A is currently active. User B taps on the avatar, and in response, the system expands a tray element within the map interface. The tray element displays User A's profile identifier "Henry," a last-active timestamp "Last active 2h ago," and a listening status indicator showing "(It Goes Like) Nanana · Listening on SPOTIFY (e.g., an individual third-party media streaming service)." The tray element also renders an add element displaying the text "Add" alongside a streaming service logo, a camera icon, and a chat icon. A streaming service identifier within the tray indicates that the media consumption activity data was retrieved from the individual streaming service.

User B selects the add element. User B has linked a second streaming service account (e.g., a different third-party media streaming service), different from the individual (first) streaming service, to User B's account on the social interaction platform. The system identifies the ISRC "NLZ542200042" from the standardized activity record, queries an internal media library of the social interaction platform, and locates a corresponding track entry matching the ISRC. From the corresponding track entry, the system resolves a platform-specific track identifier "1440825033" for the second streaming service linked to User B's account. The system transmits a write request via the second streaming service's application programming interface, using the platform-specific track identifier "1440825033," to save the track to User B's library on the second streaming service. The second streaming service returns a confirmation response. In response, the system updates the add element from a first state displaying "Add" to a second state displaying "Added," and causes display of a confirmation notification banner at the top of User B's map interface reading "Added to your Liked Songs playlist." If User A were viewing User A's own map interface, the callout element positioned proximate to User A's own avatar would display a platform-specific icon identifying the first streaming service, for example, a green circular logo with horizontal lines, rather than the generic media icon displayed to User B.

In a further example, User A closes the social interaction platform application and does not reopen it for 26 hours. During this period, User A continues to play music on the first streaming service. After 24 hours of inactivity on the social interaction platform, the backend streaming integration service ceases retrieval of media consumption activity data from the first streaming service for User A. No API requests are transmitted to the first streaming service's backend on behalf of User A, and no listening status indicator is displayed to any of User A's friends, even though User A remains actively listening on the first streaming service. When User A reopens the social interaction platform application at hour 26, the system detects that User A is actively consuming media on the first streaming service at the time of reopening. The system resumes retrieval of media consumption activity data from the first streaming service, restores User A's prior sharing configuration of "All Friends," and generates a reminder notification for display within the social interaction platform on User A's device. The reminder notification informs User A that User A's listening status indicator is visible to User A's friends and presents selectable options including an option to continue sharing and an option to activate a private listening mode. If User A selects the private listening mode option with a duration parameter of 3 hours, the system suppresses display of the listening status indicator for User A to all friends, ceases retrieval of media consumption activity data from the first streaming service for User A for 3 hours, and upon expiration of the 3-hour period, deactivates the private listening mode and resumes retrieval and display according to User A's prior sharing preference selection of "All Friends."

In a further example, the social interaction platform 510 may integrate with multiple third-party media streaming services 132 that expose media consumption activity data with different levels of latency and granularity. A first third-party media streaming service 132 may provide a real-time playback API that returns the track a user is currently playing at the moment the API is queried, along with a playback state of "playing" and a timestamp corresponding to the current playback position. A second third-party media streaming service 132 may not provide an equivalent real-time playback API, and instead may expose a "recently played" endpoint that returns a list of tracks the user has played, ordered by recency, with each entry including a track identifier, artist name, album name, and a timestamp indicating when the track was last played. The normalization module 512 handles these differences through the partner-specific interface modules. For the first third-party media streaming service 132, the partner-specific interface module maps the real-time playback response directly to the unified schema, populating the playback state field with "playing" and the event timestamp field with the current timestamp. For the second third-party media streaming service 132, the partner-specific interface module maps the most recent entry in the recently played list to the unified schema, populating the playback state field with "recently played" and the event timestamp field with the timestamp from the recently played entry. The presence logic module 514 may apply different threshold configurations when evaluating standardized activity records originating from different partner services. For example, the first threshold used to distinguish between a live listening state and a recently listened state may be set to 60 seconds for the first third-party media streaming service 132, where real-time playback data is available, and may be set to 300 seconds for the second third-party media streaming service 132, where only recently played data is available, to account for the inherent latency in the recently played endpoint.

The social interaction platform 510 may also adapt the UI text rendered within the listening status indicator based on the partner service identifier in the standardized activity record. When the standardized activity record originates from a partner that provides real-time playback data, the UI update module 516 may render text such as "Listening on [Service Name]" to indicate present-tense activity. When the standardized activity record originates from a partner that provides only recently played data, the UI update module 516 may render text such as "Listening Activity on [Service Name]" or "Recently played on [Service Name]" to reflect the approximate nature of the data without representing it as a fully live signal. This partner-aware text rendering ensures that the listening status indicator accurately represents the nature of the underlying data to the viewing user, while maintaining a consistent visual layout and interaction model across all partner integrations.

NETWORKED COMPUTING ENVIRONMENT

FIG. 1 is a block diagram showing an example digital interaction system 100 for facilitating interactions and engagements (e.g., exchanging text messages, conducting text audio and video calls, or playing games) over a network. The digital interaction system 100 includes multiple user systems 102 (e.g., user devices) and/or head-wearable apparatus 116, each of which hosts multiple applications, including an interaction client 104 and other applications 106. Each interaction client 104 is communicatively coupled, via one or more networks including a network 108 (e.g., the Internet), to other instances of the interaction client 104 (e.g., hosted on respective other user systems 102), a server system 110 and third-party servers 112, including one or more third-party media streaming services 132). An interaction client 104 can also communicate with locally hosted applications 106 using Application Programming Interfaces (APIs).

Each user system 102 may include multiple user devices, such as a mobile device 114, head-wearable apparatus 116, and a computer client device 118 that are communicatively connected to exchange data and messages.

An interaction client 104 interacts with other interaction clients 104 and with the server system 110 via the network 108. The data exchanged between the interaction clients 104 (e.g., interactions 120) and between the interaction clients 104 and the server system 110 includes functions (e.g., commands to invoke functions) and payload data (e.g., text, audio, video, or other multimedia data).

The server system 110 provides server-side functionality via the network 108 to the interaction clients 104. While certain functions of the digital interaction system 100 are described herein as being performed by either an interaction client 104 or by the server system 110, the location of certain functionality either within the interaction client 104 or the server system 110 may be a design choice. For example, it may be technically preferable to initially deploy particular technology and functionality within the server system 110 but to later migrate this technology and functionality to the interaction client 104 where a user system 102 has sufficient processing capacity.

The server system 110 supports various services and operations that are provided to the interaction clients 104. Such operations include transmitting data to, receiving data from, and processing data generated by the interaction clients 104. This data may include message content, client device information, geolocation information, digital effects (e.g., media augmentation and overlays), message content persistence conditions, entity relationship information, and live event information. Data exchanges within the digital interaction system 100 are invoked and controlled through functions available via user interfaces (UIs) of the interaction clients 104.

The third-party media streaming services 132 are external services, hosted on the third-party servers 112, that provide on-demand access to media content (e.g., music, videos, audiobooks, and so forth) for playback by users. Each third-party media streaming service 132 maintains its own user accounts, media catalogs, and playback infrastructure, and exposes one or more application programming interfaces through which external systems may retrieve data regarding user playback activity, media metadata, and library management operations. Examples of third-party media streaming services 132 include music streaming platforms, audio content platforms that provide access to podcasts or audiobooks, and/or video streaming platforms. Different third-party media streaming services 132 may expose playback activity data in different formats, with different fields, and with different latency characteristics. For example, a first third-party media streaming service 132 may provide a real-time indication of a track currently being played by a user, while a second third-party media streaming service 132 may provide a record of the most recently played track without an explicit real-time playback signal.

The server system 110 communicates with the third-party media streaming services 132 via the network 108 to retrieve media consumption activity data on behalf of users of the digital interaction system 100 who have linked their accounts on the respective third-party media streaming services 132 to their accounts on the digital interaction system 100. The server system 110 manages authentication credentials, such as OAuth access tokens, for each linked user account and uses these credentials to issue API requests to the third-party media streaming services 132 at defined polling intervals or in response to subscription-based update notifications where supported by a given third-party media streaming service 132. The server system 110 receives responses from the third-party media streaming services 132 containing partner-specific data fields describing playback activity, parses these responses using partner-specific interface modules, and converts the partner-specific data into standardized activity records according to a unified schema maintained by the server system 110. This architecture enables the server system 110 to integrate with multiple third-party media streaming services 132 concurrently and to process media consumption activity data from different third-party media streaming services 132 using a common set of downstream operations, including presence state determination, privacy rule evaluation, and rendering of listening status indicators on surfaces of the interaction client 104.

Turning now specifically to the server system 110, an Application Program Interface (API) server 122 is coupled to and provides programmatic interfaces to servers 124, making the functions of the servers 124 accessible to interaction clients 104, other applications 106, and the third-party server 112. The servers 124 are communicatively coupled to a database server 126, facilitating access to a database 128 that stores data associated with interactions processed by the servers 124. Similarly, a web server 130 is coupled to the servers 124 and provides web-based interfaces to the servers 124. To this end, the web server 130 processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

The Application Programming Interface (API) server 122 receives and transmits interaction data (e.g., commands and message payloads) between the servers 124 and the user systems 102 (and, for example, interaction clients 104 and other applications 106) and the third-party server 112. Specifically, the Application Program Interface (API) server 122 provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the interaction client 104 and other applications 106 to invoke functionality of the servers 124. The Application Program Interface (API) server 122 exposes various functions supported by the servers 124, including account registration; login functionality; the sending of interaction data, via the servers 124, from a particular interaction client 104 to another interaction client 104; the communication of media files (e.g., images or video) from an interaction client 104 to the servers 124; the settings of a collection of media data (e.g., a narrative); the retrieval of a list of friends of a user of a user system 102; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity relationship graph (e.g., the entity graph 308 in FIG. 3); the location of friends within an entity relationship graph; and opening an application event (e.g., relating to the interaction client 104).

The servers 124 host multiple systems and subsystems, described below with reference to FIG. 2.

EXTERNAL RESOURCES AND LINKED APPLICATIONS

The interaction client 104 provides a user interface that allows users to access features and functions of an external resource, such as a linked application 106, an applet, or a microservice. This external resource may be provided by a third party or by the creator of the interaction client 104. The external resource may be a full-scale application installed on the user's system 102, or a smaller, lightweight version of the application, such as an applet or a microservice, hosted either on the user's system or remotely, such as on third-party servers 112 or in the cloud. These smaller versions, which include a subset of the full application's features, may be implemented using a markup-language document and may also incorporate a scripting language and a style sheet.

When a user selects an option to launch or access the external resource, the interaction client 104 determines whether the resource is web-based or a locally installed application. Locally installed applications can be launched independently of the interaction client 104, while applets and microservices can be launched or accessed via the interaction client 104.

If the external resource is a locally installed application, the interaction client 104 instructs the user's system to launch the resource by executing locally stored code. If the resource is web-based, the interaction client 104 communicates with third-party servers 112 to obtain a markup-language document corresponding to the selected resource, which it then processes to present the resource within its user interface.

The interaction client 104 can also notify users of activity in one or more external resources. For instance, it can provide notifications relating to the use of an external resource by one or more members of a user group. Users can be invited to join an active external resource or to launch a recently used but currently inactive resource.

The interaction client 104 can present a list of available external resources to a user, allowing them to launch or access a given resource. This list can be presented in a context-sensitive menu, with icons representing different applications, applets, or microservices varying based on how the menu is launched by the user.

SYSTEM ARCHITECTURE

FIG. 2 is a block diagram illustrating further details regarding the digital interaction system 100, according to some examples. Specifically, the digital interaction system 100 is shown to comprise the interaction client 104 and the servers 124. The digital interaction system 100 embodies multiple subsystems, which are supported on the client-side by the interaction client 104 and on the server-side by the servers 124. In some examples, these subsystems are implemented as microservices. A microservice subsystem (e.g., a microservice application) may have components that enable it to operate independently and communicate with other services. Example components of microservice subsystem may include:

- Function logic: The function logic implements the functionality of the microservice subsystem, representing a specific capability or function that the microservice provides.

- API interface: Microservices may communicate with each other components through well-defined APIs or interfaces, using lightweight protocols such as REST or messaging. The API interface defines the inputs and outputs of the microservice subsystem and how it interacts with other microservice subsystems of the digital interaction system 100.
- Data storage: A microservice subsystem may be responsible for its own data storage, which may be in the form of a database, cache, or other storage mechanism (e.g., using the database server 126 and database 128). This enables a microservice subsystem to operate independently of other microservices of the digital interaction system 100.
- Service discovery: Microservice subsystems may find and communicate with other microservice subsystems of the digital interaction system 100. Service discovery mechanisms enable microservice subsystems to locate and communicate with other microservice subsystems in a scalable and efficient way.
- Monitoring and logging: Microservice subsystems may need to be monitored and logged to ensure availability and performance. Monitoring and logging mechanisms enable the tracking of health and performance of a microservice subsystem.

In some examples, the digital interaction system 100 may employ a monolithic architecture, a service-oriented architecture (SOA), a function-as-a-service (FaaS) architecture, or a modular architecture:
Example subsystems are discussed below.
An image processing system 202 provides various functions that enable a user to capture and modify (e.g., augment, annotate or otherwise edit) media content associated with a message.
A camera system 204 includes control software (e.g., in a camera application) that interacts with and controls camera hardware (e.g., directly or via operating system controls) of the user system 102 to modify real-time images captured and displayed via the interaction client 104.
A digital effect system 206 provides functions related to the generation and publishing of digital effects (e.g., media overlays) for images captured in real-time by cameras of the user system 102 or retrieved from memory of the user system 102. For example, the digital effect system 206 operatively selects, presents, and displays digital effects (e.g., media overlays such as image filters or modifications) to the interaction client 104 for the modification of real-time images received via the camera system 204 or stored images retrieved from memory of a user system 102. The digital effect system 206 can provide such functions by accessing a set of instructions associated with each respective digital effects experience and processing such instructions by video generative machine learning models in real time. The generative machine learning models can continuously process inputs and/or interactions with the rendered digital effects experiences to update presentation of the digital effects provided by the digital effects experiences. These digital effects are selected by the digital effect system 206 and presented to a user of an interaction client 104, based on a number of inputs and data, such as for example:

- Geolocation of the user system 102; and
- Entity relationship information of the user of the user system 102.

Digital effects may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. Examples of visual effects include color overlays and media overlays. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo or video) at user system 102 for communication in a message, or applied to video content, such as a video content stream or feed transmitted from an interaction client 104. As such, the image processing system 202 may interact with, and support, the various subsystems of the communication system 208, such as the messaging system 210 and the video communication system 212. A digital effect(s) application (or digital effects experience) is an application configured to provide and display these digital effects and can enable users to engage in multiplayer digital effects sessions using respective head-wearable apparatuses 116 or other user system 102. The digital effect application can be part of the application 106 (and/or interaction client 104) implemented by the user system 102 and/or the head-wearable apparatus 116. In some cases, the digital effect application or output representing the digital effect application can be rendered by a generative machine learning model by processing a set of instructions including prompts that define behavior, goals, and attributes of digital effects relative to real-world or virtual items presented in a video or image in real time. This way, rather than using SLAM or other real-time object tracking and modeling, the digital effects can be presented using fewer hardware and software resources by processing the instructions and generating outputs with the generative machine learning model. In some cases, the prompts can instruct the generative machine learning model (GenAI) to process an image or video of an avatar along with audio input and to generate a video that depicts the avatar (lips of the avatar) speaking speech provided by the audio input.
A media overlay may include text or image data that can be overlaid on top of a photograph taken by the user system 102 or a video stream produced by the user system 102. In some examples, the media overlay may be a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In further examples, the image processing system 202 uses the geolocation of the user system 102 to identify a media overlay that includes the name of a merchant at the geolocation of the user system 102. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the databases 128 and accessed through the database server 126.
The image processing system 202 provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The image processing system 202 generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.
The digital effect creation system 214 supports AR developer platforms and includes an application for content creators (e.g., artists and developers) to create and publish digital effects (e.g., AR experiences) of the interaction client 104. The digital effect creation system 214 provides a library of built-in features and tools to content creators including, for example custom shaders, tracking technology, and templates. Any functionality that is performed by the digital effect creation system 214 can be replaced and/or augmented by processing instructions with or by a generative machine learning model. In such cases, object tracking and 3D modeling components used by the digital effect creation system 214 can be omitted or skipped as the appropriate output is rendered by the generative machine learning model.
In some examples, the digital effect creation system 214 provides a merchant-based publication platform that enables merchants to select a particular digital effect associated with a geolocation via a bidding process. For example, the digital effect creation system 214 associates a media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.
A communication system 208 is responsible for enabling and processing multiple forms of communication and interaction within the digital interaction system 100 and includes a messaging system 210, an audio communication system 216, and a video communication system 212. The messaging system 210 is responsible, in some examples, for enforcing the temporary or time-limited access to content by the interaction clients 104. The messaging system 210 incorporates multiple timers that, based on duration and display parameters associated with a message or collection of messages (e.g., a narrative), selectively enable access (e.g., for presentation and display) to messages and associated content via the interaction client

104. The audio communication system 216 enables and supports audio communications (e.g., real-time audio chat) between multiple interaction clients 104. Similarly, the video communication system 212 enables and supports video communications (e.g., real-time video chat) between multiple interaction clients 104. In some cases, the communication system 208 can be used to perform the operations discussed for presenting and controlling sponsored conversation cells and sponsored content.

A user management system 218 is operationally responsible for the management of user data and profiles, and maintains entity information (e.g., stored in entity tables 306, entity graphs 308, and profile data 302, shown in FIG. 3) regarding users and relationships between users of the digital interaction system 100.

A collection management system 220 is operationally responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an "event gallery" or an "event collection." Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a "concert collection" for the duration of that music concert. The collection management system 220 may also be responsible for publishing an icon that provides notification of a particular collection to the user interface of the interaction client 104. The collection management system 220 includes a curation function that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system 220 employs machine vision (or image recognition technology) and content rules to curate a content collection automatically. In certain examples, compensation may be paid to a user to include user-generated content into a collection. In such cases, the collection management system 220 operates to automatically make payments to such users to use their content.

A map system 222 provides various geographic location (e.g., geolocation) functions and supports the presentation of map-based media content and messages by the interaction client 104. For example, the map system 222 enables the display of user icons or avatars (e.g., stored in profile data 302) on a map to indicate a current or past location of "friends" of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the digital interaction system 100 from a specific geographic location may be displayed within the context of a map at that particular location to "friends" of a specific user on a map interface of the interaction client 104. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the digital interaction system 100 via the interaction client 104, with this location and status information being similarly displayed within the context of a map interface of the interaction client 104 to selected users.

A game system 224 provides various gaming functions within the context of the interaction client 104. The interaction client 104 provides a game interface providing a list of available games that can be launched by a user within the context of the interaction client 104 and played with other users of the digital interaction system 100. The digital interaction system 100 further enables a particular user to invite other users to participate in the play of a specific game by issuing invitations to such other users from the interaction client 104. The interaction client 104 also supports audio, video, and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and supports the provision of in-game rewards (e.g., coins and items).

An external resource system 226 provides an interface for the interaction client 104 to communicate with remote servers (e.g., third-party servers 112) to launch or access external resources, e.g., applications or applets. Each third-party server 112 hosts, for example, a markup language (e.g., HTML5) based application or a small-scale version of an application (e.g., game, utility, payment, or ride-sharing application). The interaction client 104 may launch a web-based resource (e.g., application) by accessing the HTML5 file from the third-party servers 112 associated with the web-based resource. Applications hosted by third-party servers 112 are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the servers 124. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. The servers 124 host a JavaScript library that provides a given external resource access to specific user data of the interaction client 104. HTML5 is an example of technology for programming games, but applications and resources programmed based on other technologies can be used.

To integrate the functions of the SDK into the web-based resource, the SDK is downloaded by the third-party server 112 from the servers 124 or is otherwise received by the third-party server 112. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the interaction client 104 into the web-based resource.

The SDK stored on the server system 110 effectively provides the bridge between an external resource (e.g., applications 106 or applets) and the interaction client 104. This gives the user a seamless experience of communicating with other users on the interaction client 104 while also preserving the look and feel of the interaction client 104. To bridge communications between an external resource and an interaction client 104, the SDK facilitates communication between third-party servers 112 and the interaction client 104. A bridge script running on a user system 102 establishes two one-way communication channels between an external resource and the interaction client 104. Messages are sent between the external resource and the interaction client 104 via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

By using the SDK, not all information from the interaction client 104 is shared with third-party servers 112. The SDK limits which information is shared based on the needs of the external resource. Each third-party server 112 provides an HTML5 file corresponding to the web-based external resource to servers 124. The servers 124 can add a visual representation (such as a box art or other graphic) of the web-based external resource in the interaction client 104. Once the user selects the visual representation or instructs the interaction client 104 through a graphical user interface (GUI) of the interaction client 104 to access features of the web-based external resource, the interaction client 104 obtains the HTML5 file and instantiates the resources to access the features of the web-based external resource.

The interaction client 104 presents a GUI (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the interaction client 104 determines whether the launched external resource has been previously authorized to access user data of the interaction client 104. In response to determining that the launched external resource has been previously authorized to access user data of the interaction client 104, the interaction client 104 presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the interaction client 104, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the interaction client 104 slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the interaction client 104 adds the external resource to a list of authorized external resources and allows the external resource to access user data from the interaction client 104. The external resource is authorized by the interaction client 104 to access the user data under an OAuth 2 framework.

The interaction client 104 controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale applications (e.g., an application 106) are provided with access to a first type of user data (e.g., two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of applications (e.g., web-based versions of applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

An advertisement system 228 operationally enables the purchasing of advertisements by third parties for presentation to end-users via the interaction clients 104 and handles the delivery and presentation of these advertisements.

An artificial intelligence and machine learning system 230 provides a variety of services to different subsystems within the digital interaction system 100.

For example, the artificial intelligence and machine learning system 230 operates with the image processing system 202 and the camera system 204 to analyze images and extract information such as objects, text, or faces. This information can then be used by the image processing system 202 to enhance, filter, or manipulate images. The artificial intelligence and machine learning system 230 may be used by the digital effect system 206 to generate modified content and augmented reality experiences, such as adding virtual objects or animations to real-world images. The artificial intelligence and machine learning system 230 can access a set of instructions that define an individual digital effects experience. The artificial intelligence and machine learning system 230 can then process such instructions by a generative machine learning model (in some cases along with additional user supplied inputs and/or videos/images) to render an artificial video that depicts digital effects within a real-world or virtual environment defined by the instructions.

Machine learning is a field of study that gives computers the ability to learn without being explicitly programmed. The artificial intelligence and machine learning system 230 can be built using machine learning models. Machine learning (e.g., machine learning models) explores the study and construction of algorithms, also referred to herein as tools, that may learn from existing data and make predictions about new data. Such machine-learning tools operate by building a model from example training data in order to make data-driven predictions or decisions expressed as outputs or assessments. Although examples are presented with respect to a few machine-learning tools, the principles presented herein may be applied to other machine-learning tools.

In some examples, different machine-learning tools may be used. For example, Logistic Regression (LR), Naive-Bayes, Random Forest (RF), neural networks (NN), matrix factorization, and Support Vector Machines (SVM) tools may be used for classifying or scoring job postings.

Two common types of problems in machine learning are classification problems and regression problems. Classification problems, also referred to as categorization problems, aim at classifying items into one of several category values (for example, is this object an apple or an orange?). Regression algorithms aim at quantifying some items (for example, by providing a value that is a real number). The machine-learning algorithms use features for analyzing the data to generate an assessment. Each of the features is an individual measurable property of a phenomenon being observed. The concept of a feature is related to that of an explanatory variable used in statistical techniques such as linear regression. Choosing informative, discriminating, and independent features is important for the effective operation of the pattern recognition, classification, and regression. Features may be of different types, such as numeric features, strings, and graphs.

In one example, the features may be of different types and may include one or more of content, concepts, attributes, historical data, and/or user data, merely for example. The machine-learning algorithms use the training data to find correlations among the identified features that affect the outcome or assessment. In some examples, the training data includes labeled data, which is known data for one or more identified features and one or more outcomes, such as detecting communication patterns, detecting the meaning of the message, generating a summary of a message, detecting action items in messages detecting urgency in the message, detecting a relationship of the user to the sender, calculating score attributes, calculating message scores, detecting an error in an uncorrected gaze vector, etc.

With the training data and the identified features, the machine-learning tool is trained at machine-learning program training. The machine-learning tool appraises the value of the features as they correlate to the training data. The result of the training is the trained machine-learning program. When the trained machine-learning program is used to perform an assessment, new data is provided as an input to the trained machine-learning program, and the trained machine-learning program generates the assessment as output.

The machine-learning program supports two types of phases, namely a training phase and prediction phase. In training phases, supervised learning, unsupervised learning, or reinforcement learning may be used. For example, the machine-learning program (1) receives features (e.g., as structured or labeled data in supervised learning) and/or (2) identifies features (e.g., unstructured or unlabeled data for unsupervised learning) in training data. In prediction phases, the machine-learning program uses the features for analyzing query data to generate outcomes or predictions (as examples of an assessment).

In the training phase, feature engineering is used to identify features and may include identifying informative, discriminating, and independent features for the effective operation of the machine-learning program in pattern recognition, classification, and regression. In some examples, the training data includes labeled data, which is known data for pre-identified features and one or more outcomes. Each of the features may be a variable or attribute, such as individual measurable property of a process, article, system, or phenomenon represented by a data set (e.g., the training data).

In training phases, the machine-learning program uses the training data to find correlations among the features that affect a predicted outcome or assessment. With the training data and the identified features, the machine-learning program is trained during the training phase at machine-learning program training. The machine-learning program appraises values of the features as they correlate to the training data. The result of the training is the trained machine-learning program (e.g., a trained or learned model).

Further, the training phases may involve machine learning, in which the training data is structured (e.g., labeled during preprocessing operations), and the trained machine-learning program implements a relatively simple neural network capable of performing, for example, classification and clustering operations. In other examples, the training phase may involve deep learning, in which the training data is unstructured, and the trained machine-learning program implements a deep neural network that is able to perform both feature extraction and classification/clustering operations.

A neural network generated during the training phase, and implemented within the trained machine-learning program, may include a hierarchical (e.g., layered) organization of neurons. For example, neurons (or nodes) may be arranged hierarchically into a number of layers, including an input layer, an output layer, and multiple hidden layers. Each of the layers within the neural network can have one or many neurons, and each of these neurons operationally computes a small function (e.g., activation function). For example, if an activation function generates a result that transgresses a particular threshold, an output may be communicated from that neuron (e.g., transmitting neuron) to a connected neuron (e.g., receiving neuron) in successive layers. Connections between neurons also have associated weights, which defines the influence of the input from a transmitting neuron to a receiving neuron.

In some examples, the neural network may also be one of a number of different types of neural networks, including a single-layer feed-forward network, an Artificial Neural Network (ANN), a Recurrent Neural Network (RNN), a symmetrically connected neural network, an unsupervised pretrained network, a Convolutional Neural Network (CNN), a Generative Adversarial Network (GAN), and/or a Recursive Neural Network (RNN), merely for example.

During prediction phases, the trained machine-learning program is used to perform an assessment. Query data is provided as an input to the trained machine-learning program, and the trained machine-learning program generates the assessment as output, responsive to receipt of the query data.

For training the generative machine learning model to animate sponsored content characters, the training data includes pairs of audio files containing speech and corresponding video data showing synchronized lip movements and facial expressions (visemes) that match the personality and style of advertiser-specific characters. The model learns to identify key features in the speech audio, such as phonemes, timing, and emotional tone that align with the character's traits, and correlate these with appropriate character-specific animations. During the training phase, the model processes structured training data that includes labeled pairs of speech audio and corresponding facial animations. Feature engineering focuses on extracting relevant characteristics from both the audio input (such as speech patterns, pauses, and tonal variations) and the animation output (such as lip positions, facial expressions, and temporal synchronization) that maintain the character's unique personality and mannerisms.

The neural network architecture implemented for character animation includes specialized components like Generative Adversarial Networks (GANs) that can generate realistic facial animations matching the advertised character's appearance, and Recurrent Neural Networks (RNNs) that can process the temporal aspects of speech while maintaining character consistency. The model learns to generate smooth, natural-looking animations that accurately reflect both the timing and characteristics of the input speech as well as the character's distinct traits. In the prediction phase, when processing new user speech input, the trained model analyzes the audio features and generates corresponding character animations in real-time that preserve the character's unique personality. The model can handle various speech modifications, including language translation and emotional style changes, by learning correlations between different speech patterns and appropriate character-specific animation responses.

The artificial intelligence and machine learning system provides automated agent functionality that enables interactive conversations between users and an automated agent directly within the public comment threads of media content items. The system leverages multimodal content analysis and natural language processing capabilities to generate responses based on the specific context of the media content and the surrounding conversation history. This enables a shared social experience where the automated agent actively participates in public discussions, offering context-aware insights or reactions that enrich the viewing experience for the entire community without requiring private, one-on-one interactions.

The artificial intelligence and machine learning system 230 leverages natural language processing and computer vision capabilities to enable context-aware interactions where the automated agent analyzes the media content, understands ongoing conversation threads, and generates relevant responses when tagged by users in public comments. This enables users to seamlessly integrate AI assistance into their social media discussions, allowing them to ask questions about video content, request summaries or explanations, and receive AI-generated responses that become part of the persistent conversation history visible to all viewers. The automated agent maintains a consistent identity across all interactions, appearing as a unified participant in public discussions with recognizable visual indicators and attribution, while selectively determining when to respond based on safety criteria, content relevance, and conversation value to optimize resource utilization and enhance the quality of public discourse surrounding media content.

A compliance system 232 facilitates compliance by the digital interaction system 100 with data privacy and other regulations, including for example the California Consumer Privacy Act (CCPA), General Data Protection Regulation (GDPR), and Digital Services Act (DSA). The compliance system 232 comprises several components that address data privacy, protection, and user rights, ensuring a secure environment for user data. A data collection and storage component securely handles user data, using encryption and enforcing data retention policies. A data access and processing component provides controlled access to user data, ensuring compliant data processing and maintaining an audit trail. A data subject rights management component facilitates user rights requests in accordance with privacy regulations, while the data breach detection and response component detects and responds to data breaches in a timely and compliant manner. The compliance system 232 also incorporates opt-in/opt-out management and privacy controls across the digital interaction system 100, empowering users to manage their data preferences. The compliance system 232 is designed to handle sensitive data by obtaining explicit consent, implementing strict access controls and in accordance with applicable laws.

DATA ARCHITECTURE

FIG. 3 is a schematic diagram illustrating data structures 300, which may be stored in the database 128 of the server system 110, according to certain examples. While the content of the database 128 is shown to comprise multiple tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database 128 includes message data stored within a message table 304. This message data includes at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table 304, are described below with reference to FIG. 3.

An entity table 306 stores entity data, and is linked (e.g., referentially) to an entity graph 308 and profile data 302. Entities for which records are maintained within the entity table 306 may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the server system 110 stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

The entity graph 308 stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization), interest-based, or activity-based, merely for example. Certain relationships between entities may be unidirectional, such as a subscription by an individual user to digital content of a commercial or publishing user (e.g., a newspaper or other digital media outlet, or a brand). Other relationships may be bidirectional, such as a "friend" relationship between individual users of the digital interaction system 100.

Certain permissions and relationships may be attached to each relationship, and to each direction of a relationship. For example, a bidirectional relationship (e.g., a friend relationship between individual users) may include authorization for the publication of digital content items between the individual users, but may impose certain restrictions or filters on the publication of such digital content items (e.g., based on content characteristics, location data or time of day data). Similarly, a subscription relationship between an individual user and a commercial user may impose different degrees of restrictions on the publication of digital content from the commercial user to the individual user, and may significantly restrict or block the publication of digital content from the individual user to the commercial user. A particular user, as an example of an entity, may record certain restrictions (e.g., by way of privacy settings) in a record for that entity within the entity table 306. Such privacy settings may be applied to all types of relationships within the context of the digital interaction system 100, or may selectively be applied to certain types of relationships.

In some examples, the entity graph 308 and the entity table 306 also store data used by the system to control the visibility of media consumption activity among users. When a user configures a sharing preference for a listening status indicator, the system stores the sharing preference selection in association with the user's record in the entity table 306. The sharing preference selection specifies one or more visibility rules that govern which other users may view the user's listening status indicator. These visibility rules may reference relationship data stored in the entity graph 308. For example, a visibility rule specifying "all friends" causes the system to evaluate the entity graph 308 to identify all entities having a bidirectional friend relationship with the user. A visibility rule specifying "friends I share location with" causes the system to identify the subset of friend entities for which the user has an existing location-sharing permission recorded in the entity table 306 or the entity graph 308. A visibility rule specifying a custom list of individually selected friends causes the system to store and reference a list of entity identifiers associated with the user's record. Prior to rendering a listening status indicator for a given user on a surface presented to a viewing user, the system evaluates whether the viewing user satisfies the applicable visibility rules by querying the entity graph 308 and the entity table 306.

The profile data 302 stores multiple types of profile data about a particular entity. The profile data 302 may be selectively used and presented to other users of the digital interaction system 100 based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data 302 includes, for example, a username, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the digital interaction system 100, and on map interfaces displayed by interaction clients 104 to other users. The collection of avatar representations may include "status avatars," which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

In some examples, the profile data 302 for an individual user also stores data associated with linked third-party media streaming service accounts and media consumption activity. When a user completes an OAuth-based authorization flow to link a third-party media streaming service account to the user's account on the digital interaction system 100, the profile data 302 stores an association between the user's entity identifier and the linked streaming service, including an access token, a refresh token, and a partner identifier indicating which third-party media streaming service 132 the user has linked. The profile data 302 may also store a current or most recent standardized activity record for the user, including fields such as a track identifier, an artist identifier, an album identifier, a playback state, a timestamp, and a session identifier. This standardized activity record is updated by the server system 110 as new media consumption activity data is retrieved from the third-party media streaming service 132 and converted according to the unified schema. The profile data 302 further stores a private listening mode indicator and, where applicable, a duration parameter specifying when the private listening mode is set to expire. When the system renders a map interface, a profile interface, or another surface for a viewing user, the system accesses the profile data 302 of each friend of the viewing user to retrieve the current standardized activity record, the sharing preference selection, and the private listening mode indicator, and uses these data to determine whether and how to render a listening status indicator for each friend.

Additionally, the profile data 302 may store an internal media library association for each track referenced in a standardized activity record. When media consumption activity data is retrieved from a third-party media streaming service 132 and normalized, the system extracts an international standard recording code (ISRC) or other standardized track identifier from the standardized activity record and queries an internal media library maintained within the database 128 to locate a corresponding track entry. If a match is found, the profile data 302 or a related table stores the association between the standardized track identifier and one or more platform-specific track identifiers corresponding to different third-party media streaming services 132. This association enables the system to resolve a platform-specific track identifier for a viewing user's linked streaming service when the viewing user selects an add element to save a track to the viewing user's own library, even when the viewing user's linked streaming service differs from the streaming service from which the media consumption activity data was originally retrieved.

Where the entity is a group, the profile data 302 for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

The database 128 also stores digital effect data, such as overlays or filters, in a digital effect table 310. The digital effect data is associated with and applied to videos (for which data is stored in a video table 312) and images (for which data is stored in an image table 314).

Filters, in some examples, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the interaction client 104 when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the interaction client 104, based on geolocation information determined by a Global Positioning System (GPS) unit of the user system 102.

Another type of filter is a data filter, which may be selectively presented to a sending user by the interaction client 104 based on other inputs or information gathered by the user system 102 during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a user system 102, or the current time.

Other digital effect data (e.g., instructions that define one or more digital effects experiences) that may be stored within the image table 314 include augmented reality content items (e.g., corresponding to augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

A collections table 316 stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a narrative or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table 306). A user may create a "personal collection" in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the interaction client 104 may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal narrative.

A collection may also constitute a "live collection," which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a "live collection" may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the interaction client 104, to contribute content to a particular live collection. The live collection may be identified to the user by the interaction client 104, based on his or her location.

A further type of content collection is known as a "location collection," which enables a user whose user system 102 is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location collection may employ a second degree of authentication to verify that the end-user belongs to a specific organization or other entity (e.g., is a student on the university campus).

As mentioned above, the video table 312 stores video data that, in some examples, is associated with messages for which records are maintained within the message table 304. Similarly, the image table 314 stores image data associated with messages for which message data is stored in the entity table 306. The entity table 306 may associate various digital effects from the digital effect table 310 with various images and videos stored in the image table 314 and the video table 312.

The databases 128 also include a list of digital effects experiences along with their respective sets of instructions (that define their operation) and/or code that is executed by tracking systems to provide outputs of the digital effects experiences.

DATA COMMUNICATIONS ARCHITECTURE

FIG. 4 is a schematic diagram illustrating a structure of a message 400, according to some examples, generated by an interaction client 104 for communication to a further interaction client 104 via the servers 124. The content of a particular message 400 is used to populate the message table

304 stored within the database 128, accessible by the servers 124. Similarly, the content of a message 400 is stored in memory as "in-transit" or "in-flight" data of the user system 102 or the servers 124. A message 400 is shown to include the following example components:

- Message identifier 402: a unique identifier that identifies the message 400.
- Message text payload 404: text, to be generated by a user via a user interface of the user system 102, and that is included in the message 400.
- Message image payload 406: image data, captured by a camera component of a user system 102 or retrieved from a memory component of a user system 102, and that is included in the message 400. Image data for a sent or received message 400 may be stored in the image table 314.
- Message video payload 408: video data, captured by a camera component or retrieved from a memory component of the user system 102, and that is included in the message 400. Video data for a sent or received message 400 may be stored in the video table 312.
- Message audio payload 410: audio data, captured by a microphone or retrieved from a memory component of the user system 102, and that is included in the message 400.
- Message digital effect data 412: digital effect data (e.g., filters, stickers, or other annotations or enhancements) that represents digital effects to be applied to message image payload 406, message video payload 408, or message audio payload 410 of the message 400. Digital effect data for a sent or received message 400 may be stored in the digital effect table 310.
- Message duration parameter 414: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload 406, message video payload 408, message audio payload 410) is to be presented or made accessible to a user via the interaction client 104.
- Message geolocation parameter 416: geolocation data (e.g., latitudinal, and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter 416 values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload 406, or a specific video in the message video payload 408).
- Message collection identifier 418: identifier values identifying one or more content collections (e.g., "stories" identified in the collections table 316) with which a particular content item in the message image payload 406 of the message 400 is associated. For example, multiple images within the message image payload 406 may each be associated with multiple content collections using identifier values.
- Message tag 420: each message 400 may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload 406 depicts an animal (e.g., a lion), a tag value may be included within the message tag 420 that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.
- Message sender identifier 422: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system 102 on which the message 400 was generated and from which the message 400 was sent.
- Message receiver identifier 424: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system 102 to which the message 400 is addressed.

The contents (e.g., values) of the various components of message 400 may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload 406 may be a pointer to (or address of) a location within an image table 314. Similarly, values within the message video payload 408 may point to data stored within a video table 312, values stored within the message digital effect data 412 may point to data stored in a digital effect table 310, values stored within the message collection identifier 418 may point to data stored in a collections table 316, and values stored within the message sender identifier 422 and the message receiver identifier 424 may point to user records stored within an entity table 306.

FIG. 5 illustrates a diagram 504 depicting an architecture for retrieving media consumption activity data from an external media service 506 and processing that data through a series of modules within a social interaction platform 510 to render a listening status indicator on a user device display 518, in accordance with some examples. The diagram 504 shows the flow of data from the external media service 506 through an API 508 into the social interaction platform 510, where the data passes through a normalization module 512, a presence logic module 514, and a UI update module 516 before being presented on the user device display 518. The components shown in FIG. 5 may operate in conjunction with the server system 110, the interaction client 104, the third-party media streaming services 132, and the database 128 described with reference to FIGS. 1-4.

The external media service 506 represents one or more third-party media streaming services 132 as described with reference to FIG. 1. The external media service 506 maintains user accounts, media catalogs, and playback infrastructure, and exposes the API 508 through which the social interaction platform 510 may retrieve data regarding a user's playback activity. The API 508 may correspond to a REST-based or other programmatic interface provided by the external media service 506. The social interaction platform 510 communicates with the external media service 506 via the API 508 over the network 108 to retrieve media consumption activity data associated with a first user who has linked an account on the external media service 506 to the first user's account on the social interaction platform 510.

Prior to retrieving media consumption activity data, the social interaction platform 510 may execute an OAuth-based authorization flow with the external media service 506 on behalf of the first user. The authorization flow may comprise transmitting an authorization request to the external media service 506 via the API 508 and receiving an access token in response to the first user granting consent. The social interaction platform 510 may store the access token in association with an account of the first user, for example within the profile data 302 in the database 128 as described with reference to FIG. 3. The social interaction platform 510 may detect that the access token has expired or been invalidated by the external media service 506 and may execute a token refresh operation to obtain a replacement access token. In response to determining that the token refresh operation has failed, the social interaction platform 510 may transmit a notification to the first user indicating that a connection to the external media service 506 is no longer active.

The social interaction platform 510 retrieves media consumption activity data from the external media service 506 via the API 508. The retrieval may occur via polling at defined intervals, for example every 15 seconds, or through subscription-based updates where supported by the external media service 506. The media consumption activity data received from the external media service 506 may include partner-specific fields such as a partner track identifier, an artist field, an album field, a playback state field indicating whether the user is actively playing a track, and a timestamp indicating when the playback event occurred or was last updated. Different instances of the external media service 506 may expose media consumption activity data in different formats, with different fields, and with different latency characteristics. For example, a first external media service 506 may provide a real-time indication of a track currently being played, while a second external media service 506 may provide a record of the most recently played track without an explicit real-time playback signal.

The media consumption activity data received from the external media service 506 passes into the normalization module 512 within the social interaction platform 510. The normalization module 512 converts the media consumption activity data into a standardized activity record according to

a unified schema. The unified schema may correspond to the unified media activity structure 604 described with reference to FIG. 6. The normalization module 512 parses a response received from the API 508 of the external media service 506 to extract one or more partner-specific data fields. The normalization module 512 maps each of the one or more partner-specific data fields to a corresponding field of the unified schema. The normalization module 512 populates the standardized activity record with values derived from the mapping.
The unified schema may comprise one or more fields selected from a group consisting of a user identifier, a track identifier, an artist identifier, an album identifier, a playback state, a timestamp, a session identifier, and a partner identifier. For example, the normalization module 512 may receive a response from a first external media service 506 containing a partner track identifier "3WOhcATHxK1X," an artist field "Peggy Gou," an album field "I Hear You," a playback state field indicating "playing," and a timestamp of "2026-03-12T14:32:07Z." The normalization module 512 may map these partner-specific fields to the corresponding fields of the unified schema, generating a standardized activity record comprising a user identifier "userA12345," a track identifier corresponding to the international standard recording code "NLZ542200042," an artist identifier "Peggy Gou," an album identifier "I Hear You," a playback state of "playing," a timestamp of "2026-03-12T14:32:07Z," a session identifier "sess8837a," and a partner identifier "streamingservice_1."
Where the social interaction platform 510 integrates with multiple instances of the external media service 506, the normalization module 512 may use a partner-specific interface module associated with each respective external media service 506. Each partner-specific interface module maps one or more fields of the media consumption activity data received from the respective external media service 506 to corresponding fields of the unified schema. The standardized activity record generated for each external media service 506 includes a partner identifier distinguishing that external media service 506 from other instances of the external media service 506. This architecture enables the normalization module 512 to produce standardized activity records in a common format regardless of which external media service 506 provided the underlying data, and enables downstream modules within the social interaction platform 510 to operate on the unified schema without dependency on the data format or interface conventions of any particular external media service 506.
The standardized activity record generated by the normalization module 512 passes to the presence logic module 514. The presence logic module 514 determines, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user. The presence state indicates a status of the first user's media consumption on the external media service 506. The presence logic module 514 may compare the timestamp of the standardized activity record against a current time to compute an elapsed duration. In response to determining that the elapsed duration is below a first threshold and the playback state indicates active playback, the presence logic module 514 may classify the presence state as a live listening state. In response to determining that the elapsed duration is above the first threshold and below a second threshold, the presence logic module 514 may classify the presence state as a recently listened state. In response to determining that the elapsed duration exceeds the second threshold, the presence logic module 514 may classify the presence state as an inactive state and may suppress display of a listening status indicator for the first user.
For example, if the presence logic module 514 receives a standardized activity record with a timestamp of "2026-03-12T14:32:07Z" and the current time is "2026-03-12T14:32:15Z," the presence logic module 514 computes an elapsed duration of 8 seconds. If the first threshold is 60 seconds and the playback state indicates "playing," the presence logic module 514 classifies the presence state as a live listening state. If the same standardized activity record were evaluated at a current time of "2026-03-12T14:37:07Z," yielding an elapsed duration of 300 seconds, and the second threshold were 600 seconds, the presence logic module 514 may classify the presence state as a recently listened state. If the elapsed duration exceeded the second threshold, the presence logic module 514 may classify the presence state as an inactive state.
The presence logic module 514 may also evaluate the first user's sharing preference selection prior to permitting display of the listening status indicator. The sharing preference selection may specify one or more visibility rules for the listening status indicator. The one or more visibility rules may comprise at least one rule selected from a group consisting of sharing with all friends of the first user, sharing with a subset of friends with whom the first user shares location data, and sharing with a custom list of one or more individually selected friends. The presence logic module 514 may access the entity graph 308 and the entity table 306 stored in the database 128, as described with reference to FIG. 3, to evaluate whether a second user satisfies the one or more visibility rules specified by the sharing preference selection. The presence logic module 514 may permit display of the listening status indicator to the second user only in response to determining that the second user satisfies the one or more visibility rules.
The presence logic module 514 may also process a selection of a private listening mode received from the first user. The selection may specify a duration parameter indicating one of a fixed time period, such as 3 hours or 24 hours, or an indefinite period. In response to receiving the selection of the private listening mode, the presence logic module 514 may suppress display of the listening status indicator for the first user to all friends of the first user regardless of the sharing preference selection. The social interaction platform 510 may cease retrieval of the media consumption activity data from the external media service 506 for the first user for the duration specified by the duration parameter. In response to determining that the duration parameter has elapsed, the social interaction platform 510 may automatically or based on confirmation from the user deactivate the private listening mode and resume retrieval of the media consumption activity data and display of the listening status indicator according to the sharing preference selection.
The presence logic module 514 may also interact with an application activity monitor 710, as described with reference to FIG. 7. The application activity monitor 710 may monitor an application activity signal associated with the first user on the social interaction platform 510. The application activity signal may indicate a last time the first user opened or interacted with the social interaction platform 510. The presence logic module 514 may determine that a duration since the last time the first user opened or interacted with the social interaction platform 510 exceeds an inactivity threshold, for example 24 hours. In response to determining that the duration exceeds the inactivity threshold, the social interaction platform 510 may cease retrieval of the media consumption activity data from the external media service 506 for the first user. In response to detecting a subsequent opening of the social interaction platform 510 by the first user, the social interaction platform 510 may resume retrieval of the media consumption activity data from the external media service 506 and restore a prior sharing configuration associated with the first user.
Upon detecting that the first user has reopened the social interaction platform 510 after the duration exceeded the inactivity threshold, the presence logic module 514 may determine that the first user is actively consuming media on the external media service 506 at the time of the reopening. The social interaction platform 510 may generate a reminder notification for display on the device of the first user. The reminder notification may inform the first user that the listening status indicator is visible to one or more friends of the first user. The reminder notification may present one or more selectable options comprising at least an option to continue sharing and an option to activate a private listening mode.
Once the presence logic module 514 determines that a presence state permits display and that the second user satisfies the applicable visibility rules, the standardized activity record and the presence state are passed to the UI update module 516. The UI update module 516 coordinates rendering of a listening status indicator on one or more surfaces of the social interaction platform 510 presented on the user device display 518. The user device display 518 corresponds to a display of a user system 102 as described with reference to FIG. 1, such as a display of the mobile device 114 or the computer client device 118.
The UI update module 516 may render the listening status indicator within a map interface on the user device display 518. The map interface may correspond to a surface provided by the map system 222 as described with reference to FIG. 2. The UI update module 516 may determine a geographic location of the first user based on one or more location signals received from a device of the first user, render an avatar representing the first user at a

position on the map interface corresponding to the geographic location, and generate a callout element positioned proximate to the avatar. The callout element may comprise a track name and an artist name extracted from the standardized activity record. The UI update module 516 may render, within the callout element, a selectable element that, upon selection by the second user, initiates navigation to a content page associated with a track identified in the standardized activity record.

The UI update module 516 may also render the listening status indicator within other surfaces of the social interaction platform 510. For example, the UI update module 516 may render the listening status indicator within a profile interface of the first user as a persistent indicator element displaying at least a track name and an artist name from the standardized activity record. The UI update module 516 may render the listening status indicator within a contextual tray element that is displayed in response to a selection interaction by the second user on the map interface. The UI update module 516 may render the listening status indicator within a content page associated with a media track identified in the standardized activity record.

When the UI update module 516 renders the listening status indicator on the map interface, the UI update module 516 may determine whether the viewing user to whom the map interface is being presented corresponds to the first user or to a user other than the first user. In response to determining that the viewing user corresponds to the first user, the UI update module 516 may render, within the callout element or the listening status indicator, a platform-specific icon identifying the external media service 506 from which the media consumption activity data was retrieved. In response to determining that the viewing user corresponds to a user other than the first user, the UI update module 516 may render a generic media icon in place of the platform-specific icon. The generic media icon may be visually distinct from the platform-specific icon and may not identify a particular external media service 506.

The UI update module 516 may render one or more interactive elements within the listening status indicator or within a tray element associated with the listening status indicator. The one or more interactive elements may comprise at least the add song button 714, as described with reference to FIG. 7. The add song button 714 may be configured to initiate a save operation for a track identified in the standardized activity record. Upon detecting a selection of the add song button 714 by the second user, the social interaction platform 510 may identify, from the standardized activity record, a track identifier associated with the track. The social interaction platform 510 may transmit, via a second API associated with a streaming service linked to an account of the second user, a write request to save the track to a library of the second user on the streaming service linked to the account of the second user. In response to receiving a confirmation that the write request was processed, the UI update module 516 may update the add song button 714 from a first state indicating an available save action to a second state indicating that the track has been saved, and may cause display of a confirmation notification on the user device display 518.

Where the streaming service linked to the account of the second user is different from the external media service 506 associated with the first user, the social interaction platform 510 may extract an international standard recording code from the standardized activity record associated with the track. The social interaction platform 510 may match the international standard recording code to a corresponding track entry in an internal media library maintained within the database 128. The social interaction platform 510 may resolve, from the corresponding track entry, a platform-specific track identifier for the streaming service linked to the account of the second user. The social interaction platform 510 may transmit the write request to the streaming service linked to the account of the second user using the platform-specific track identifier resolved from the corresponding track entry.

The UI update module 516 may also handle scenarios in which the second user has not linked a streaming service account to the social interaction platform 510. Upon detecting an interaction by the second user with the listening status indicator, such as a selection of the add song button 714, the UI update module 516 may present a connection prompt to the second user, such as the account linking prompt 708 described with reference to FIG. 7. The connection prompt may comprise one or more selectable options each corresponding to a different external media service 506 available for linking. The social interaction platform 510 may initiate an authorization flow with an external media service 506 selected by the second user from the one or more selectable options. The user action logger 716, as described with reference to FIG. 7, may log the interaction and a surface from which the connection prompt was presented as a conversion event.

The social interaction platform 510 may present the connection prompt from one or more entry points within the social interaction platform 510. The one or more entry points may comprise at least two of a map tray element, a content page associated with a media track, a settings interface, a profile indicator element, or a feed banner element. The user action logger may record, for each presentation of the connection prompt, an entry point identifier indicating which of the one or more entry points triggered the presentation. The social interaction platform 510 may compute a conversion rate for each entry point identifier based on a ratio of completed streaming service linkings to presentations of the connection prompt from that entry point. The social interaction platform 510 may adjust a frequency or a priority of presenting the connection prompt at each of the one or more entry points based on the computed conversion rates.

The UI update module 516 may also support deep linking functionality. Upon detecting a selection by the second user of the listening status indicator or a portion thereof, the social interaction platform 510 may generate a deep link comprising one or more parameters identifying a track indicated in the standardized activity record. The one or more parameters may comprise at least a partner-specific track identifier compatible with the external media service 506 or a streaming service linked to the second user. The social interaction platform 510 may determine whether a native application associated with a target streaming service is installed on the user device display 518. In response to determining that the native application is installed, the social interaction platform 510 may transmit the deep link to the native application to cause the native application to navigate to a content view associated with the track. In response to determining that the native application is not installed, the social interaction platform 510 may redirect the second user to a web-based interface or an application store listing associated with the target streaming service.

The social interaction platform 510 may also render listening status indicators within a Sounds Topic Page surface. A Sounds Topic Page is a content page within the social interaction platform 510 that is associated with a particular media track and aggregates content items, such as user-generated video content, that incorporate or reference that media track. When a first user's standardized activity record identifies a track that has a corresponding Sounds Topic Page within the social interaction platform 510, the UI update module 516 may render a listening status indicator or a reference to the first user's listening activity on the Sounds Topic Page. The Sounds Topic Page may display the track title, artist name, a count of content items that use the track, and one or more content thumbnails. The Sounds Topic Page may also render an add song button that enables a viewing user to save the track to the viewing user's linked streaming service library. When the viewing user selects the add song button on the Sounds Topic Page, the social interaction platform 510 performs the same save operation described above, including the cross-platform track resolution using the ISRC unique media code when the viewing user's linked streaming service differs from the streaming service associated with the track's origin. The Sounds Topic Page may also serve as one of the multiple onboarding entry points from which the social interaction platform 510 may present a connection prompt to a user who has not yet linked a streaming service account.

The social interaction platform 510 may also render a listening status indicator or a track reference within a content item detail view. When a viewing user encounters a content item, such as a short-form video, that uses a particular media track as its audio component, the content item detail view may display the track title and artist name, along with a selectable element that navigates the viewing user to the corresponding Sounds Topic Page. The content item detail view may also render an add song button that enables the viewing user to save the track directly from the content item detail view without navigating to the Sounds Topic Page. This integration enables the social interaction platform 510 to surface media discovery opportunities across content consumption surfaces in addition to the map interface and profile interface.

The social interaction platform 510 may also support audio preview functionality in connection with the listening status indicator. When a viewing user interacts with a callout element on the map interface or with a listening status indicator on another surface, the social interaction platform 510 may initiate playback of a preview clip of the track identified in the standardized activity record. The preview clip may be a segment of the track, such as a 30-second excerpt, that is retrieved from the internal media library of the social interaction platform 510 or from the third-party media streaming service 132 via the API 508. The preview clip may be played within the social interaction platform 510 without requiring the viewing user to navigate to an external streaming application. In some examples, the third-party media streaming service 132 may provide the preview clip as part of its API response, and the social interaction platform 510 may render a playback control within the callout element or the tray element to allow the viewing user to play, pause, or stop the preview. The audio preview functionality enables the viewing user to sample the track before deciding whether to save it to the viewing user's library or to navigate to the full track on the streaming service via a deep link.

The architecture shown in the diagram 504 of FIG. 5 may operate with multiple instances of the external media service 506 concurrently. For example, a first user may link a first external media service 506 while a third user may link a second, different external media service 506. The normalization module 512 may use respective partner-specific interface modules to convert media consumption activity data from each external media service 506 into standardized activity records according to the same unified schema. The presence logic module 514 may determine presence states for both the first user and the third user using the unified schema regardless of which external media service 506 provided the media consumption activity data. The UI update module 516 may render listening status indicators for both users on the same surface of the social interaction platform 510 presented on the user device display 518, with each listening status indicator comprising at least a portion of the respective standardized activity record.

FIG. 6 illustrates a unified media activity structure 604, which represents a data structure used to store a standardized activity record generated by the normalization module 512 described with reference to FIG. 5, in accordance with some examples. The unified media activity structure 604 defines the unified schema according to which media consumption activity data retrieved from one or more instances of the external media service 506 via the API 508 is converted into a common format. The fields shown in the unified media activity structure 604 enable downstream components of the social interaction platform 510, including the presence logic module 514 and the UI update module 516, to process and render media consumption information without dependency on the data format or interface conventions of any particular third-party media streaming service 132 as described with reference to FIG. 1.

The unified media activity structure 604 includes a Unified Schema Definition field, which identifies the version or type of the schema used to structure the standardized activity record. The Unified Schema Definition field may allow the system to distinguish between different versions of the schema as the system evolves to support additional media types or additional fields over time. The social interaction platform 510 may reference the Unified Schema Definition field when processing standardized activity records to confirm that the record conforms to the expected format before passing it to the presence logic module 514 or the UI update module 516.

The unified media activity structure 604 includes a User Identifier field, which stores a value that uniquely identifies the user of the social interaction platform 510 whose media consumption activity is represented by the standardized activity record. The User Identifier field may correspond to an entity identifier maintained in the entity table 306 within the database 128 as described with reference to FIG. 3. For example, a User Identifier value of "userA12345" may correspond to a record in the entity table 306 for a user named "Henry." The User Identifier field enables the system to associate the standardized activity record with the correct user account when evaluating sharing preferences stored in the profile data 302, when querying friend relationships in the entity graph 308, and when rendering a listening status indicator on a surface presented to a viewing user.

The unified media activity structure 604 includes a Track Name and Artist field, which stores values identifying the media track and the associated artist or creator. These values may be extracted by the normalization module 512 from partner-specific fields in the response received from the API 508 of the external media service 506. For example, the normalization module 512 may extract a track name of "(It Goes Like) Nanana" and an artist name of "Peggy Gou" from a response received from a first external media service 506 and populate the Track Name and Artist field accordingly. The Track Name and Artist field provides the data from which the UI update module 516 generates the text displayed within a callout element on a map interface, within a persistent indicator on a profile interface, or within a tray element, as described with reference to FIG. 5.

The unified media activity structure 604 includes a Playback State and Progress field, which stores a value indicating the current or most recent status of the user's media playback on the external media service 506. The Playback State and Progress field may take values such as "playing," "paused," or "recently played." The presence logic module 514 uses the Playback State and Progress field in conjunction with the Event Timestamp field to determine the presence state for the user. For example, if the Playback State and Progress field indicates "playing" and the Event Timestamp field indicates a time within a first threshold of the current time, the presence logic module 514 may classify the presence state as a live listening state. If the Playback State and Progress field indicates "paused" or if the Event Timestamp exceeds the first threshold, the presence logic module 514 may classify the presence state as a recently listened state. Different instances of the external media service 506 may expose playback state information with varying levels of granularity; for example, a first external media service 506 may provide a real-time "playing" indication, while a second external media service 506 may provide only a "recently played" record, and the normalization module 512 maps these varying representations into the Playback State and Progress field of the unified media activity structure 604.

The unified media activity structure 604 includes an Event Timestamp field, which stores a value indicating when the playback event occurred or was last updated by the external media service 506. The Event Timestamp field is used by the presence logic module 514 to compute an elapsed duration by comparing the Event Timestamp against a current time. The computed elapsed duration, together with the Playback State and Progress field, determines whether the presence state is classified as a live listening state, a recently listened state, or an inactive state. The Event Timestamp field may also be used by the UI update module 516 to render temporal labels within a listening status indicator, such as a last-active timestamp displayed within a tray element on a map interface.

The unified media activity structure 604 includes a Partner Service Identifier field and an ISRC Unique Media Code field. The Partner Service Identifier field stores a value that identifies the particular third-party media streaming service 132 from which the media consumption activity data was retrieved, such as "streamingservice1" or "streamingservice2." This field enables the system to distinguish standardized activity records originating from different instances of the external media service 506 and to render the appropriate platform-specific icon or generic media icon within the listening status indicator based on whether the viewing user is the same as the consuming user. The ISRC Unique Media Code field stores an international standard recording code or other standardized track identifier associated with the media track. The ISRC Unique Media Code field enables cross-platform track identification; for example, when a viewing user who has linked a second external media service 506 selects an add element to save a track that was originally retrieved from a first external media service 506, the system may use the ISRC Unique Media Code to query an internal media library within the database 128, locate a corresponding track entry, and resolve a platform-specific track identifier for the viewing user's linked streaming service. This cross-platform resolution allows the system to transmit a write request to the viewing user's streaming service using the resolved platform-specific track identifier, even when the viewing user's streaming service differs from the streaming service associated with the consuming user.

FIG. 7 illustrates a diagram 704 depicting the interaction between user-facing interface components and backend monitoring and logging components within the social interaction platform 510, in accordance with some examples. The diagram 704 shows how a listening status indicator 706, an account linking prompt 708, a tray element 712, and an add song button 714 are presented within a user interface on a device of a viewing user, and how user interactions with these components are recorded by a user action logger 716. The diagram 704 also shows an application activity monitor 710 that operates alongside the user interface components to track the activity state of a user on the social interaction platform 510. The components shown in FIG. 7 may operate in conjunction with the normalization module 512, the presence logic module 514, and the UI update module 516 described with reference to FIG. 5, and may reference data stored in the unified media activity structure 604 described with reference to FIG. 6.

The diagram 704 depicts a user interface rendered on a device of a second user. Within the user interface, the listening status indicator 706 is displayed in an upper portion of the screen. The listening status indicator 706 presents media consumption activity information associated with a first user, such as a track name, an artist name, and a playback state, derived from a standardized activity record generated according to the unified media activity structure 604. The listening status indicator 706 may be rendered by the UI update module 516 within one or more surfaces of the social interaction platform 510, including a map interface, a profile interface, or a content page. The listening status indicator 706 may include text such as a track title and artist name, and may also include a visual representation of the playback state, such as an animated waveform or a static icon indicating that the first user is listening to media. The specific visual layout and content of the listening status indicator 706 may vary based on partner-specific display requirements associated with the external media service 506 from which the media consumption activity data was retrieved.

Below the listening status indicator 706, the diagram 704 shows the account linking prompt 708. The account linking prompt 708 is a modal or overlay element that the social interaction platform 510 presents to the second user in response to detecting that the second user has not yet linked a streaming service account to the social interaction platform 510. The account linking prompt 708 may be triggered when the second user interacts with the listening status indicator 706 or with the add song button 714 while no streaming service account is linked. The account linking prompt 708 comprises one or more selectable options, each corresponding to a different third-party media streaming service 132 available for linking, such as a first streaming service button and a second streaming service button. The account linking prompt 708 may also include a "Not Now" option allowing the second user to dismiss the prompt without linking a streaming service. A dashed arrow in the diagram 704 connects the account linking prompt 708 to the application activity monitor 710, indicating that the presentation of the account linking prompt 708 and the second user's response to it may be tracked by the application activity monitor 710 and recorded by the user action logger 716.

The diagram 704 shows the tray element 712 positioned in a lower portion of the user interface. The tray element 712 is an expandable interface component that the social interaction platform 510 displays in response to detecting a selection by the second user of an avatar or a callout element on a map interface. The tray element 712 may comprise a profile identifier of the first user, a last-active timestamp associated with the first user, the listening status indicator 706, and a streaming service identifier indicating the third-party media streaming service 132 from which the media consumption activity data was retrieved. The streaming service identifier is shown in the diagram 704 as a service identifier icon rendered within the tray element 712, such as a logo corresponding to the first user's linked streaming service. The tray element 712 provides the second user with additional context about the first user's media consumption activity beyond what is visible in the callout element on the map interface.

Within the tray element 712, the diagram 704 shows the add song button 714. The add song button 714 is an interactive element configured to initiate a save operation for a track identified in the standardized activity record associated with the first user. The add song button 714 may be rendered in a first state displaying text such as "Add" when the track has not yet been saved to the second user's library, and may transition to a second state displaying text such as "Added" after the save operation completes. Upon detecting a selection of the add song button 714 by the second user, the social interaction platform 510 identifies a track identifier from the standardized activity record, transmits a write request via an API associated with a streaming service linked to the second user's account, and upon receiving a confirmation that the write request was processed, updates the add song button 714 to the second state and causes display of a confirmation notification. Where the second user has not yet linked a streaming service account, selection of the add song button 714 may trigger presentation of the account linking prompt 708 instead of initiating the save operation. The placement and prominence of the add song button 714 relative to other interactive elements within the tray element 712 may vary across the three map treatments; for example, in a first treatment the add song button 714 may be rendered as a primary call-to-action element, in a second treatment the add song button 714 may be rendered alongside a chat element as co-equal interactive elements, and in a third treatment the add song button 714 may be rendered within the callout element on the map interface rather than or in addition to being rendered within the tray element 712.

The diagram 704 shows the application activity monitor 710 as a component positioned outside the user interface. The application activity monitor 710 monitors an application activity signal associated with the first user on the social interaction platform 510. The application activity signal indicates a last time the first user opened or interacted with the social interaction platform 510. The presence logic module 514 described with reference to FIG. 5 may use data from the application activity monitor 710 to determine whether a duration since the first user's last interaction with the social interaction platform 510 exceeds an inactivity threshold, such as 24 hours. In response to determining that the duration exceeds the inactivity threshold, the social interaction platform 510 may cease retrieval of media consumption activity data from the external media service 506 for the first user. The application activity monitor 710 may also detect when the first user reopens the social interaction platform 510 after the inactivity threshold has been exceeded, and may trigger the social interaction platform 510 to resume retrieval of media consumption activity data and to generate a reminder notification informing the first user that the listening status indicator 706 is visible to one or more friends of the first user.

At the bottom of the diagram 704, the user action logger 716 is shown as a data store that receives and records user interaction events from the user interface components. The user action logger 716 logs interactions such as selections of the add song button 714, presentations and dismissals of the account linking prompt 708, tray element 712 open and close events, and deep link tap-through events. For each logged interaction, the user action logger 716 may record an entry point identifier indicating which surface or component of the social interaction platform 510 triggered the interaction, such as a map tray element, a content page, a settings interface, a profile indicator element, or a feed banner element. The social interaction platform 510 may use data recorded by the user action logger 716 to compute a conversion rate for each entry point identifier based on a ratio of completed streaming service linkings to presentations of the account linking prompt 708 from that entry point. The social interaction platform 510 may adjust a frequency or a priority of presenting the account linking prompt 708 at each entry point based on the computed conversion rates. The user action logger 716 may also record streaming intent events, such as when the second user taps on the listening status indicator 706 and is redirected to the external media service 506 via a deep link, enabling the social interaction platform 510 to measure downstream discovery impact.

FIG. 8 illustrates a user interface 804 depicting a map interface 806 rendered on a device of a user of the social interaction platform 510, in accordance with some examples. The user interface 804 shows a map-based view with geographic features such as streets, terrain contours, and landmarks. Overlaid on the map interface 806 are several user interface elements related to media consumption activity sharing, including a listening status indicator 808 and privacy settings 810. The user interface 804 corresponds to one of the surfaces on which the UI update module 516, described with reference to FIG. 5, may render listening activity information derived from a standardized activity record stored according to the unified media activity structure 604 described with reference to FIG. 6.

In an upper portion of the user interface 804, a sound integration element is displayed. The sound integration element includes a label reading "SOUND INTEGRATION" and an icon, indicating a feature area within the social interaction platform 510 that relates to media content and audio functionality. The sound integration element may correspond to a navigation element or an entry point through which a user may access content pages

associated with media tracks, such as a Sounds Topic Page. The sound integration element may serve as one of the multiple entry points from which the social interaction platform 510 may present a connection prompt, as described with reference to the account linking prompt 708 in FIG. 7.
The map interface 806 occupies a central region of the user interface 804 and displays a geographic area with rendered streets, intersections, and topographic features. Positioned on the map interface 806 is an avatar icon representing a user. The avatar icon is rendered at a position on the map interface 806 corresponding to a geographic location of the user, as determined by one or more location signals received from the user's device. The avatar icon may correspond to a user-selected avatar representation stored in the profile data 302 within the database 128, as described with reference to FIG. 3. The map interface 806 may be provided by the map system 222 described with reference to FIG. 2.
Above the avatar icon on the map interface 806, the listening status indicator 808 is displayed as a callout element positioned proximate to the avatar. The listening status indicator 808 includes text reading "LISTENING TO," a track title field displaying "TRACK TITLE," and an artist name field displaying "ARTIST NAME." These fields correspond to values extracted from the Track Name and Artist field of the unified media activity structure 604 described with reference to FIG. 6. The listening status indicator 808 also includes an icon to the left of the text, which may represent either a platform-specific icon or a generic media icon depending on whether the viewing user corresponds to the user whose media consumption activity is being displayed. As described with reference to FIG. 5, when the viewing user corresponds to the user whose activity is shown, the UI update module 516 may render a platform-specific icon identifying the external media service 506 from which the media consumption activity data was retrieved. When the viewing user is a different user, the UI update module 516 may render a generic media icon that does not identify a particular external media service 506. The listening status indicator 808 may be selectable; upon selection by a viewing user, the social interaction platform 510 may initiate navigation to a content page associated with the track identified in the standardized activity record, or may expand a tray element displaying additional information as described with reference to FIG. 7.
Below the map interface 806, the user interface 804 displays a prompt element containing the text "LINK MUSIC SERVICE" and a "CONNECT" button. This prompt element corresponds to a connection prompt that the social interaction platform 510 presents to a user who has not yet linked a streaming service account. The prompt element functions in a manner consistent with the account linking prompt 708 described with reference to FIG. 7. Upon selection of the "CONNECT" button, the social interaction platform 510 may present one or more selectable options corresponding to different third-party media streaming services 132 available for linking, and may initiate an OAuth-based authorization flow with a streaming service selected by the user. The placement of the connection prompt below the map interface 806, where the user can see the listening status indicator 808 of a friend displayed on the map, illustrates the contextual onboarding approach, in which the user first observes the value of the feature by seeing a friend's listening activity before being prompted to connect a streaming service.
At the bottom of the user interface 804, the privacy settings 810 are displayed. The privacy settings 810 include a label reading "PRIVACY SETTINGS" and a sub-label reading "SHARE LOCATION." The privacy settings 810 correspond to the sharing preference configuration described with reference to the presence logic module 514 in FIG. 5 and the share permission dialog 1106 described with reference to FIG. 11. The privacy settings 810 may provide access to controls through which the user may specify one or more visibility rules for the listening status indicator 808, such as sharing with all friends, sharing with a subset of friends with whom the user shares location data, or sharing with a custom list of individually selected friends. The privacy settings 810 may also provide access to a listen privately option, as described with reference to the listen privately option 1206 in FIG. 12, through which the user may temporarily or indefinitely suppress display of the listening status indicator 808 to all friends. The co-location of the privacy settings 810 with the map interface 806 within the user interface 804 reflects the relationship between location sharing and listening activity sharing within the social interaction platform 510, where both types of presence information may be displayed on the map interface 806 and may be subject to overlapping or independent visibility rules as configured by the user.
The user interface 804 shown in FIG. 8 represents a generalized view of the map-based surface on which the listening status indicator 808 may be rendered. The specific visual layout, button placement, and interactive elements within the user interface 804 may vary across the three map treatments (T1, T2, T3) described with reference to FIG. 7. For example, in a first treatment, an add song button may be rendered as a primary call-to-action element within a tray element below the map. In a second treatment, the add song button may be rendered alongside a chat element as co-equal interactive elements. In a third treatment, an add element may be rendered within the callout element of the listening status indicator 808 on the map interface 806 itself, rather than within a separate tray element. The social interaction platform 510 may assign users to experimental cohorts to evaluate which treatment drives the highest rate of streaming service connections, song saves, and deep-link tap-through behavior, as recorded by the user action logger 716 described with reference to FIG. 7.
FIG. 9 illustrates a first user interface 908 and a second user interface 910 depicting two views of a map interface within the social interaction platform 510, in accordance with some examples. The first user interface 908 shows an annotated breakdown of the visual components that compose the map-based listening activity surface, and the second user interface 910 shows a rendered implementation of that surface as it may appear on a device of a viewing user. The user interfaces shown in FIG. 9 correspond to the map interface 806 described with reference to FIG. 8 and illustrate how the UI update module 516, described with reference to FIG. 5, renders media consumption activity data derived from the unified media activity structure 604, described with reference to FIG. 6, within a map-based surface provided by the map system 222, described with reference to FIG. 2.
The first user interface 908 presents an annotated view identifying the constituent elements of the map-based listening activity surface. The annotations identify the following components: a settings element positioned in an upper region of the interface, a user avatar and pet element rendered on the map at a geographic position corresponding to the first user's location, a nearby venue label displayed proximate to the user avatar, reaction icons displayed in a row below the map area, a user profile card displayed in a lower portion of the interface, a navigation bar at the bottom of the interface, a current location indicator, a music player widget, a current location label, and a user profile element.
The settings element corresponds to a settings interface through which a user may access the privacy settings 810 described with reference to FIG. 8, including sharing preference controls and the listen privately option 1206 described with reference to FIG. 12. The user avatar and pet element represents a graphical avatar of the first user, rendered at a position on the map interface corresponding to the first user's geographic location as determined by one or more location signals received from the first user's device. The avatar may correspond to a user-selected avatar representation stored in the profile data 302 within the database 128, as described with reference to FIG. 3. The pet element represents a companion avatar displayed alongside the user avatar, which may be a customizable graphical element associated with the first user's profile.
The nearby venue label displays the name of a geographic point of interest proximate to the first user's location on the map, such as a restaurant, shop, or landmark. The nearby venue label may be rendered by the map system 222 based on geographic data associated with the first user's location.
The reaction icons are a set of selectable emoji or icon elements displayed in a horizontal row below the map area. The reaction icons may allow a viewing user to send a reaction to the first user in response to viewing the first user's location or listening activity on the map. A plus icon is displayed adjacent to the reaction icons, which may allow the viewing user to access additional reaction options.
The music player widget corresponds to a callout element positioned proximate to the user avatar on the map. The music player widget displays a track name and an artist name extracted from the standardized activity record stored according to the unified media activity structure 604. For example, the music player widget may display the text "(It Goes Like)..." and "Peggy Gou," corresponding to the Track Name and Artist field of the unified media activity structure 604. The music player widget may also include an icon identifying the external media service 506 from which the media

consumption activity data was retrieved. A selectable arrow or navigation element within the music player widget may, upon selection by the viewing user, initiate navigation to a content page associated with the track identified in the standardized activity record.

The user profile card is displayed in a lower portion of the first user interface 908 and corresponds to the tray element 712 described with reference to FIG. 7. The user profile card displays the first user's display name, such as "Henry," a last-active timestamp such as "Last active 2h ago," and a listening status indicator showing the track title, artist name, and a label indicating the streaming service, such as "Listening on Spotify." The user profile card also renders one or more interactive elements. These interactive elements include an add song button corresponding to the add song button 714 described with reference to FIG. 7, which is configured to initiate a save operation for the track identified in the standardized activity record. The interactive elements further include a car icon, a camera icon, and a chat icon. The add song button may display a streaming service logo alongside the text "Add" in a first state, and may transition to display the text "Added" in a second state upon completion of a save operation.

The current location indicator and the current location label identify the geographic area displayed on the map, such as "Time Square" and "71°F," providing geographic and environmental context for the first user's location.

The navigation bar at the bottom of the first user interface 908 includes a set of icons for navigating between surfaces of the social interaction platform 510, such as a location icon for the map interface, a chat icon for the messaging system 210, a camera icon for the camera system 204, and additional icons for accessing other features of the social interaction platform 510.

The second user interface 910, shown in the lower portion of FIG. 9, presents a rendered view of the map-based listening activity surface as it may appear on a device of a viewing user. The second user interface 910 shows a map of a geographic area corresponding to Times Square, New York, with streets, intersections, and landmarks rendered by the map system 222. The map displays multiple user avatars at positions corresponding to the geographic locations of respective users, as well as labels for nearby venues such as "Bond 45 Italian Kitchen & Bar," "Los Tacos No. 1 Popular with Friends," "Oh, Mary! Lyceum Theatre," and "Margaritaville - Times Square."

The first user's avatar is rendered on the map at a position corresponding to the intersection of 47th Street and 7th Avenue. A companion pet avatar is rendered alongside the first user's avatar. A label reading "Henry now" is displayed below the first user's avatar, indicating the first user's display name and that the first user is currently active. Above the first user's avatar, a callout element displays the text "(It Goes Like)..." and "Peggy Gou," along with a streaming service icon and a selectable navigation arrow. This callout element corresponds to the music player widget identified in the first user interface 908 and to the listening status indicator 808 described with reference to FIG. 8.

In the lower portion of the second user interface 910, a tray element is displayed. The tray element shows the first user's profile picture, the display name "Henry," the last-active timestamp "Last active 2h ago," and the listening status indicator "(It Goes Like) Nanana · Listening on Spotify" with a selectable arrow. Below the listening status indicator, the tray element renders a row of interactive elements including a car icon, an added song button displaying a streaming service logo and the text "Added," a camera icon, and a chat icon. The added song button corresponds to the add song button 714 described with reference to FIG. 7 and is configured to indicate that the song has been added to the user's streaming service.

At the top of the second user interface 910, a notification banner reads "Added to your Liked Songs playlist." accompanied by a heart icon. This notification banner corresponds to the confirmation notification that the UI update module 516 causes to be displayed on the viewing user's device in response to receiving a confirmation that a write request to save a track to the viewing user's library on a linked streaming service was processed, as described with reference to FIG. 5. The notification banner indicates that the viewing user previously selected the add song button and that the track was saved to the viewing user's library on the linked streaming service.

The second user interface 910 also displays additional user avatars on the map at positions corresponding to the geographic locations of other users within the viewing user's social graph. Profile pictures are rendered for other users, such as a user located near "Oh, Mary! Lyceum Theatre" and a user in the upper-left region of the map labeled with a location name. These additional avatars demonstrate that the map interface may simultaneously display listening status indicators and location information for multiple friends of the viewing user.

The map interface shown in the second user interface 910 may also display listening status indicators in a zoomed-out view. When the map interface is displayed at a zoom level where individual user avatars are not rendered at full size, the social interaction platform 510 may render a compact representation of the user's listening activity. In some examples, the compact representation may take the form of an actionmoji, which is a small animated or static avatar icon that incorporates a visual indicator of the user's current activity. When the first user is actively consuming media on a linked third-party media streaming service 132, the actionmoji rendered for the first user on the zoomed-out map view may include a visual element indicating media playback, such as a musical note icon, headphone icon, or an animated waveform overlay rendered adjacent to or superimposed on the actionmoji. The actionmoji representation enables the map interface to convey listening activity information to a viewing user even when the map is displayed at a scale where the full callout element and tray element are not visible. When the viewing user zooms into the map to a level where the first user's avatar is rendered at full size, the social interaction platform 510 transitions from the actionmoji representation to the full callout element displaying the track name, artist name, and selectable navigation element as described above.

The map interface may also support interaction with listening status indicators through reaction icons. As shown in the first user interface 908, a set of reaction icons is displayed in a horizontal row below the map area, adjacent to a plus icon. The reaction icons may include emoji or icon elements such as an eyes icon, a cake icon, a surprised face icon, and a heart icon. A viewing user may select one of the reaction icons to send a reaction to the first user in response to viewing the first user's listening activity or location on the map. The social interaction platform 510 may transmit the selected reaction to the first user as a notification or as a message within the messaging system 210 described with reference to FIG. 2. Selection of the plus icon may present additional reaction options or may allow the viewing user to compose a custom response. The reaction icons provide a lightweight interaction mechanism that enables the viewing user to acknowledge or respond to the first user's listening activity without composing a full message or navigating away from the map interface.

The two views presented in FIG. 9 together illustrate the relationship between the annotated component layout of the map-based listening activity surface and the rendered implementation of that surface. The first user interface 908 identifies each component by function, while the second user interface 910 shows how those components appear when populated with data from the unified media activity structure 604 and rendered by the UI update module 516 on the user device display 518.

FIG. 10 illustrates a first user interface state and a second user interface state depicting an onboarding flow for connecting a third-party media streaming service to the social interaction platform 510, in accordance with some examples. The first user interface state, shown on the left portion of FIG. 10, presents a map interface on a device of a second user prior to the second user having linked a streaming service account. The second user interface state, shown in the right portion of FIG. 10, presents a connection prompt that is displayed in response to the second user interacting with an element on the map interface. The user interface states shown in FIG. 10 correspond to the account linking prompt 708 described with reference to FIG. 7 and illustrate one of the multiple onboarding entry points through which the social interaction platform 510 may present a connection prompt, as described with reference to the user action logger 716 in FIG. 7.

The first user interface state displays a map interface rendered on the device of the second user. The map interface shows a geographic area with rendered streets, intersections, and landmarks, consistent with the map interface 806 described with reference to FIG. 8 and the map interface provided by the map system 222 described with reference to FIG. 2. Multiple user avatars are rendered on the map at positions corresponding to the geographic locations of respective users within the second user's social graph. Each avatar represents a friend of the second user whose location is shared on the map.

In the lower-right region of the first user interface state, a user profile card is displayed. The user profile card shows a display name, such as "USER NAME" and "USER A," identifying a first user whose media consumption activity is being surfaced. Below the display name, an add song button 1006 is rendered. The add song button 1006 corresponds to the add song button 714 described with reference to FIG. 7 and is configured to initiate a save operation for a track identified in a standardized activity record associated with the first user. In the first user interface state, the add song button 1006 is displayed in a first state indicating an available save action. A search bar is rendered in the upper portion of the first user interface state, and a navigation element is displayed in the upper-right region.

In the context of the first user interface state, the second user has not yet linked a streaming service account to the social interaction platform 510. When the second user selects the add song button 1006 or another interactive element associated with the first user's listening activity, the social interaction platform 510 detects that the second user has not linked a streaming service account. In response to detecting this condition, the social interaction platform 510 transitions to the second user interface state.

The second user interface state displays the same map interface in the background, with the user avatars and geographic features remaining visible. Overlaid on the map interface is a connection prompt (e.g., connect your preferred music app 1008). The connection prompt is a modal or overlay element that occupies a portion of the screen and presents the second user with options for linking a streaming service account to the social interaction platform 510. The connection prompt corresponds to the account linking prompt 708 described with reference to FIG. 7.

The connection prompt includes a title area containing text such as "CONNECT YOUR PREFERRED MUSIC APP" and descriptive text explaining that the second user may save songs from the social interaction platform 510 directly to a preferred music application upon connecting. Below the descriptive text, the connection prompt presents a first streaming service 1010 button and a second streaming service 1012 button. The first streaming service 1010 button is labeled with text such as "CONNECT FIRST THIRD-PARTY STREAMING SERVICE BUTTON" and corresponds to a first third-party media streaming service 132, such as a first music streaming platform. The second streaming service 1012 button is labeled with text such as "CONNECT SECOND THIRD-PARTY STREAMING SERVICE BUTTON" and corresponds to a second third-party media streaming service 132, such as a second music streaming platform. Each button includes an icon associated with the respective streaming service. The connection prompt also includes a "NOT NOW" option, allowing the second user to dismiss the prompt without linking a streaming service.

An arrow in FIG. 10 connects the add song button 1006 in the first user interface state to the connection prompt in the second user interface state, indicating the transition that occurs when the second user selects the add song button 1006 without having a linked streaming service account. This transition illustrates the contextual onboarding mechanism described with reference to the user action logger 716 in FIG. 7, in which the social interaction platform 510 presents the connection prompt in response to a user interaction with a listening status indicator or an associated interactive element, rather than requiring the user to navigate to a separate settings interface.

Upon the second user selecting one of the first streaming service 1010 button or the second streaming service 1012 button, the social interaction platform 510 initiates an OAuth-based authorization flow with the selected third-party media streaming service 132, as described with reference to FIG. 5. The authorization flow comprises transmitting an authorization request to the selected streaming service and receiving an access token in response to the second user granting consent. The social interaction platform 510 stores the access token in association with the second user's account, for example within the profile data 302 in the database 128 as described with reference to FIG. 3. Upon completion of the authorization flow, the social interaction platform 510 may return the second user to the map interface and may execute the save operation that the second user originally initiated by selecting the add song button 1006, so that the track identified in the standardized activity record associated with the first user is saved to the second user's library on the newly linked streaming service without requiring the second user to re-initiate the save action.

The user action logger 716, described with reference to FIG. 7, logs the interaction that triggered the presentation of the connection prompt, including the surface from which the connection prompt was presented. In the scenario depicted in FIG. 10, the entry point identifier recorded by the user action logger 716 corresponds to the map tray element, because the second user interacted with the add song button 1006 rendered within the user profile card on the map interface. The social interaction platform 510 may use data recorded by the user action logger 716 to compute conversion rates for this entry point relative to other entry points, such as a content page associated with a media track, a settings interface, a profile indicator element, or a feed banner element, as described with reference to FIG. 7. The social interaction platform 510 may adjust the frequency or priority of presenting the connection prompt at each entry point based on the computed conversion rates.

The connection prompt shown in FIG. 10 may also be presented from entry points other than the map interface. For example, the social interaction platform 510 may present the connection prompt when the second user navigates to a Sounds Topic Page and selects an add song element on that page, when the second user accesses a settings interface and selects a music integration option, when the second user taps on a profile indicator element displaying a friend's listening activity, or when the second user interacts with a feed banner element on a Friends Feed surface. The visual layout and content of the connection prompt may remain consistent across these entry points, while the entry point identifier recorded by the user action logger 716 varies to reflect the surface from which the prompt was triggered.

The connection prompt may also include, at the bottom of the modal, a disclaimer informing the second user that the second user's favorite tracks, listening history, and other music activity may be used to enhance the second user's experience on the social interaction platform 510, and that the second user may modify sharing preferences at any time in a settings interface under a music and now playing section.

Following completion of the account linking flow initiated from the connection prompt, the social interaction platform 510 may present a share permission dialog, such as the share permission dialog 1106 described with reference to FIG. 11, prompting the second user to configure sharing preferences for the second user's own listening activity. This sequential flow, from attempting to save a friend's song, to linking a streaming service, to configuring sharing preferences, illustrates the multi-step onboarding sequence through which the social interaction platform 510 converts a viewing user into both a consumer and a sharer of listening activity data.

FIG. 11 illustrates a user interface 1104 depicting a share permission dialog 1106 overlaid on a map interface within the social interaction platform 510, in accordance with some examples. The user interface 1104 shows the share permission dialog 1106 that the social interaction platform 510 presents to a user after the user has completed an account linking flow with a third-party media streaming service 132, as described with reference to FIG. 10. The share permission dialog 1106 enables the user to configure sharing preferences that control which friends within the user's social graph may view the user's listening status indicator. The user interface 1104 corresponds to the privacy settings 810 described with reference to FIG. 8 and illustrates the sharing preference configuration performed by the presence logic module 514 described with reference to FIG. 5.

In the upper portion of the user interface 1104, a notification banner is displayed. The notification banner includes a heart icon and text reading "ADDED TO LIKED SONGS PLAYLIST" followed by a label "NOTIFICATION BANNER." The notification banner corresponds to the confirmation notification that the UI update module 516 causes to be displayed on a viewing user's device in response to receiving a confirmation that a write request to save a track to the viewing user's library on a linked streaming service was processed, as described with reference to FIG. 7 and FIG. 9. The presence of the notification banner in the upper portion of the user interface 1104 indicates that the user interface 1104 may be presented in a state where the user has recently completed both an account linking flow and a song save operation, and is now being prompted to configure sharing preferences for the user's own listening activity.

Below the notification banner, a portion of the map interface is visible. The map interface corresponds to the map interface 806 described with reference to FIG. 8 and the map interface shown in the second user interface 910 described with reference to FIG. 9. The map interface is rendered behind the share permission dialog 1106, indicating that the share permission dialog 1106 is presented as a modal or overlay element that partially obscures the map while allowing the user to see the map context in which the listening activity sharing feature operates.

The share permission dialog 1106 occupies a lower portion of the user interface 1104 and is presented as a card or sheet element. At the top of the share permission dialog 1106, a title reads "SHARE ACTIVE LISTENING WITH FRIENDS?" Below the title, descriptive text reads "LET YOUR FRIENDS SEE WHAT YOU'RE CURRENTLY LISTENING TO ON SPOTIFY THROUGH THE SNAP MAP AND OTHER SURFACES." This descriptive text informs the user that the sharing preference being configured applies to the listening status indicator that is displayed on the map interface and on other surfaces of the social interaction platform 510, as described with reference to the multiple surfaces in FIG. 5 and FIG. 8. The reference to "SPOTIFY" in the descriptive text corresponds to the third-party media streaming service 132 that the user has linked; in other implementations, this text may reference a different streaming service name corresponding to the service the user linked during the account linking flow described with reference to FIG. 10.

Below the descriptive text, the share permission dialog 1106 presents a set of visibility rule options. Each option is accompanied by a radio button or selection indicator, indicating that the options are mutually exclusive and the user may select one option at a time.

The first option is an all friends option 1108, labeled "ALL FRIENDS." The all friends option 1108 corresponds to a visibility rule specifying that the listening status indicator associated with the user is to be displayed to all friends within the user's social graph, as maintained in the entity graph 308 within the database 128 described with reference to FIG. 3. When the all friends option 1108 is selected, the presence logic module 514 evaluates any viewing user who has a friend relationship with the sharing user as satisfying the visibility rule, and the UI update module 516 renders the listening status indicator to all such viewing users.

The second option is a selected friends option 1110, labeled "FRIENDS I SHARE LOCATION WITH" and accompanied by a list of friend names such as "JOHN, ALICE, JOE, MARK + 12 MORE..." The selected friends option 1110 corresponds to a visibility rule specifying that the listening status indicator is to be displayed only to the subset of friends with whom the user shares location data on the map interface. This option links the listening activity sharing permissions to the existing location sharing permissions maintained by the map system 222 described with reference to FIG. 2. When the selected friends option 1110 is selected, the presence logic module 514 evaluates whether a viewing user is within the set of friends with whom the sharing user has an active location sharing relationship, and only renders the listening status indicator to viewing users who satisfy this condition. In the user interface 1104 shown in FIG. 11, the selected friends option 1110 is shown with a checkmark or filled radio button, indicating that this option is the currently selected option.

Below the selected friends option 1110, a third option is displayed, labeled "ONLY THESE FRIENDS..." This option corresponds to a visibility rule specifying that the listening status indicator is to be displayed only to a custom list of one or more individually selected friends. When this option is selected, the social interaction platform 510 may present a friend selection interface allowing the user to search for and select individual friends from the user's social graph. The selected friends are stored in association with the user's account in the profile data 302 within the database 128. When the presence logic module 514 evaluates whether a viewing user satisfies the visibility rule, it checks whether the viewing user appears in the custom list. This option provides the user with the most granular control over which friends may view the user's listening activity.

Below the visibility rule options, the share permission dialog 1106 presents two action buttons. An allow button 1112 is labeled "ALLOW BUTTON." Selection of the allow button 1112 confirms the user's sharing preference selection and stores the selected visibility rule in association with the user's account on the social interaction platform 510. Upon selection of the allow button 1112, the social interaction platform 510 begins surfacing the user's listening activity to friends who satisfy the selected visibility rule. The presence logic module 514 applies the stored visibility rule when determining whether to render the listening status indicator for the user on any surface presented to a viewing user.

A not now button 1114 is labeled "NOT NOW BUTTON." Selection of the not now button 1114 dismisses the share permission dialog 1106 without configuring a sharing preference. When the user selects the not now button 1114, the social interaction platform 510 does not begin sharing the user's listening activity with any friends. The user's account linking to the third-party media streaming service 132 remains intact, meaning the user may still save songs from friends' listening status indicators to the user's own library on the linked streaming service, but the user's own listening activity is not surfaced to other users. The user may later configure sharing preferences by navigating to the settings interface, such as the settings interface shown in the user interface 1204 described with reference to FIG. 12.

At the bottom of the share permission dialog 1106, a disclaimer text is displayed. The disclaimer reads "YOUR FAVORITE TRACKS, LISTENING HISTORY, AND OTHER MUSIC ACTIVITIES CAN ENHANCE YOUR EXPERIENCE ON SNAPCHAT. YOU CAN MODIFY YOUR SHARING PREFERENCES ANYTIME IN THE SETTINGS UNDER MUSIC AND NOW PLAYING." This disclaimer informs the user that the user's music activity data may be used by the social interaction platform 510 to enhance the user's experience, and that the sharing preferences configured in the share permission dialog 1106 are not permanent and may be modified at any time through the settings interface. The reference to "MUSIC AND NOW PLAYING" in the disclaimer corresponds to the settings section shown in the user interface 1204 described with reference to FIG. 12, where the user may access the listen privately option 1206, the connected streaming service indicator 1208, and the connect additional streaming service option 1210.

The share permission dialog 1106 shown in FIG. 11 may be presented at multiple points within the user's interaction with the social interaction platform 510. As described with reference to FIG. 10, the share permission dialog 1106 may be presented immediately after the user completes the account linking flow initiated from the connection prompt 1008. The share permission dialog 1106 may also be presented when the user first opens the social interaction platform 510 after linking a streaming service account through the settings interface, or when the user returns to the social interaction platform 510 after the 24-hour inactivity threshold described with reference to the application activity monitor 710 in FIG. 7 has been exceeded and the user's listening activity sharing has been paused. In the latter case, the share permission dialog 1106 may function as a reminder notification, presenting the user with the option to continue sharing according to the prior sharing preference selection or to activate a private listening mode, as described with reference to the listen privately option 1206 in FIG. 12.

The user action logger 716 described with reference to FIG. 7 records the user's interaction with the share permission dialog 1106, including which visibility rule option the user selected and whether the user selected the allow button 1112 or the not now button 1114. This data enables the social interaction platform 510 to measure the rate at which users who complete the account linking flow proceed to enable listening activity sharing, and to identify which visibility rule option users most frequently select. These metrics inform the optimization of the onboarding flow and the presentation of the share permission dialog 1106 across different surfaces and entry points within the social interaction platform 510.

FIG. 12 illustrates a user interface 1204 depicting a settings interface for managing streaming service connections and listening activity sharing preferences within the social interaction platform 510, in accordance with some examples. The user interface 1204 is accessible from a settings menu of the social interaction platform 510 and provides a centralized location through which a user may manage linked streaming service accounts, configure listening activity visibility, and activate a private listening mode. The user interface 1204 corresponds to the privacy settings 810 described with reference to FIG. 8 and provides persistent access to the sharing preference controls that are initially presented in the share permission dialog 1106 described with reference to FIG. 11. The settings shown in the user interface 1204 are stored in association with the user's account in the profile data 302 within the database 128 described with reference to FIG. 3, and are enforced by the presence logic module 514 described with reference to FIG. 5.

At the top of the user interface 1204, a title reads "Music and Now Playing" with a back navigation arrow, indicating that this interface is a sub-section within the broader settings hierarchy of the social interaction platform 510. Below the title, descriptive text reads "Link a music service for a more personalized app experience," informing the user that connecting a streaming service account enables additional features within the social interaction platform 510.

Below the descriptive text, the user interface 1204 displays a connected streaming service indicator 1208. The connected streaming service indicator 1208 shows the name and icon of a first streaming service that the user has linked to the social interaction platform 510. In the example shown in FIG. 12, the connected streaming service indicator 1208 displays "First Streaming Service" alongside an icon corresponding to the linked service. To the right of the connected streaming service indicator 1208, a dismiss or disconnect icon is displayed, represented as an "X" symbol. Selection of this icon initiates a disconnection flow for the linked streaming service. In response to selection of this icon, the social interaction platform 510 may present a confirmation dialog informing the user that friends will no longer be able to see what the user is listening to, and providing "Disconnect" and "Cancel" options. Upon confirmation, the social interaction platform 510 revokes the stored access token associated with the linked streaming service, ceases retrieval of media consumption activity data from that service, and removes the listening status indicator for the user from all surfaces. The connected streaming service indicator 1208 transitions to an unlinked state, and the user interface 1204 updates to display a connection option in place of the linked service indicator.

Below the connected streaming service indicator 1208, the user interface 1204 displays a connect additional streaming service option 1210. The connect additional streaming service option 1210 is labeled "Connect to Second Streaming Service" alongside an icon corresponding to a second third-party media streaming service 132 and a navigation arrow. Selection of the connect additional streaming service option 1210 initiates an account linking flow for the second streaming service. When a user who already has one streaming service linked selects the option to connect a different streaming service, the social interaction platform 510 may present a confirmation dialog informing the user that connecting the new service will automatically disconnect the previously linked service. This mutual exclusivity constraint, in which only one streaming service may be linked at a time, is reflected in the user interface 1204 by the presence of a single connected streaming service indicator 1208 and a single connect additional streaming service option 1210. The social interaction platform 510 enforces this constraint by revoking the access token for the previously linked service and initiating an OAuth-based authorization flow for the newly selected service, as described with reference to FIG. 5 and FIG. 10.

Below the streaming service connection section, the user interface 1204 displays a "Now Playing" section. A heading reads "Now Playing" followed by descriptive text reading "The song you're currently playing will be visible to these friends." This section corresponds to the sharing preference configuration that the user initially set through the share permission dialog 1106 described with reference to FIG. 11, and provides persistent access to modify those preferences.

Within the "Now Playing" section, the user interface 1204 displays a listen privately option 1206. The listen privately option 1206 is presented as a toggle element with an associated label reading "Listen Privately" and descriptive text reading "When enabled, your friends can't see your current playing track." The listen privately option 1206 corresponds to the private listening mode. When the user activates the listen privately option 1206, the social interaction platform 510 suppresses display of the listening status indicator for the user to all friends regardless of the sharing preference selection configured below. The social interaction platform 510 also ceases retrieval of media consumption activity data from the third-party media streaming service for the user while the listen privately option 1206 is active.

Selection of the listen privately option 1206 may present a duration selection interface offering the user a choice among multiple duration parameters, such as "for 3 hours," "for 24 hours," or "Indefinitely." When the user selects a fixed time period, the social interaction platform 510 activates the private listening mode for the specified duration and displays the expiration time within the settings interface, such as "Enabled until 12:41PM" or "Enabled until tomorrow at 9:41AM." Upon expiration of the duration parameter, the social interaction platform 510 deactivates the private listening mode, resumes retrieval of media consumption activity data from the third-party media streaming service, and resumes display of the listening status indicator according to the sharing preference selection configured in the visibility rule options below. When the user selects "Indefinitely," the private listening mode remains active until the user manually deactivates it by toggling the listen privately option 1206 off.

Below the listen privately option 1206, the user interface 1204 displays a set of visibility rule options for controlling which friends may view the user's listening activity. These options correspond to the visibility rule options presented in the share permission dialog 1106 described with reference to FIG. 11, and provide the user with persistent access to modify the selection.

A first option is labeled "My Friends" and corresponds to the all friends option 1108 described with reference to FIG. 11. When selected, the presence logic module 514 renders the listening status indicator to all friends within the user's social graph as maintained in the entity graph 308 within the database 128.

A second option is labeled "Only Friends I share location with..." and is accompanied by a list of friend names such as "John, Alice, Joe, Mark + 12 more..." This option corresponds to the selected friends option 1110 described with reference to FIG. 11. When selected, the presence logic module 514 evaluates whether a viewing user is within the set of friends with whom the sharing user has an active location sharing relationship on the map interface provided by the map system 222. In the user interface 1204 shown in FIG. 12, this option is shown with a filled selection indicator, indicating that it is the currently active option. A third option is labeled "Only These Friends..." and corresponds to the custom list option described with reference to FIG. 11. When selected, the social interaction platform 510 presents a friend selection interface allowing the user to search for and select individual friends.

The user interface 1204 shown in FIG. 12 serves as the centralized management surface for all aspects of the listening activity sharing feature. From this single interface, the user may perform the following operations: link a new streaming service account by selecting the connect additional streaming service option 1210; disconnect an existing streaming service account by selecting the dismiss icon on the connected streaming service indicator 1208; activate or deactivate the private listening mode by toggling the listen privately option 1206; select a duration for the private listening mode; and modify the visibility rules controlling which friends may view the user's listening activity. Changes made through the user interface 1204 are stored in association with the user's account in the profile data 302 and are applied by the presence logic module 514 when evaluating whether to render the listening status indicator for the user on any surface presented to a viewing user.

The user interface 1204 also serves as one of the multiple onboarding entry points from which the social interaction platform 510 may present a connection prompt to a user who has not yet linked a streaming service account. When the user navigates to the user interface 1204 without having linked any streaming service, the connected streaming service indicator 1208 and the connect additional streaming service option 1210 are both displayed as connection options, each labeled with the name of a respective third-party media streaming service 132 and a "Connect" label. The "Now Playing" section and the listen privately option 1206 may not be displayed until the user completes an account linking flow, because the sharing preference controls are not applicable until a streaming service is linked. Upon completion of the account linking flow, the user interface 1204 updates to display the connected streaming service indicator 1208 for the newly linked service, and the "Now Playing" section with the listen privately option 1206 and visibility rule options becomes visible. The user action logger 716 described with reference to FIG. 7 records the entry point identifier as the settings interface when a user initiates an account linking flow from the user interface 1204. The user interface 1204 is accessible from the main settings menu of the social interaction platform 510.

FIG. 13 illustrates a flowchart 1304 depicting a process for generating and rendering a listening status indicator within the social interaction platform 510, in accordance with some examples. The flowchart 1304 shows a sequence of operations performed by the social interaction platform 510 to retrieve media consumption activity data from a third-party media streaming service 132, convert the data into a standardized format, determine a presence state, and render a listening status indicator on a surface of the social interaction platform 510. The flowchart 1304 corresponds to the data flow described with reference to FIG. 5, in which data passes from the external media service 506 through the API 508 to the normalization module 512, the presence logic module 514, and the UI update module 516 for rendering on the user device display 518.

At operation 1306, the process begins with a start indicator generation step. This step represents the initiation of the listening status indicator generation process for a first user of the social interaction platform 510. The process may be initiated in response to one or more triggering conditions, such as the first user opening the social interaction platform 510 on the first user's device, a second user opening a map interface and navigating to a geographic area where the first user is located, a second user selecting an avatar or profile element associated with the first user, or a periodic polling cycle reaching its next scheduled retrieval time. The start indicator generation step may also be initiated in response to the social interaction platform 510 detecting that the first user has reopened the application after a period of inactivity that exceeded the 24-hour inactivity threshold described with reference to the application activity monitor 710 in FIG. 7 and the listen privately option described with reference to FIG. 12. In this case, the social interaction platform 510 resumes the indicator generation process after having previously ceased retrieval of media consumption activity data during the inactivity period.

At operation 1308, the system retrieves media activity via an API. This operation corresponds to the retrieval of media consumption activity data from the third-party media streaming service 132 via the application programming interface described with reference to FIG. 5. The social interaction platform 510 transmits a request to the API of the third-party media streaming service 132 associated with the first user's linked streaming account. The request includes authentication credentials, such as an access token obtained through the OAuth-based authorization flow described with reference to FIG. 10 and stored in association with the first user's account in the profile data 302 within the database 128 described with reference to FIG. 3. The third-party media streaming service 132 returns a response containing media consumption activity data in a partner-specific format. The response may include partner-specific fields such as a partner track identifier, an artist name, an album name, a playback state field indicating whether the first user is actively playing a track or has recently played a track, and a timestamp indicating when the playback event occurred or was last updated. For example, the response from a first streaming service may include fields such as a partner track identifier "3WOhcATHxK1X," an artist field "Peggy Gou," an album field "I Hear You," a playback state field indicating "playing," and a timestamp of "2026-03-12T14:32:07Z." The retrieval may occur via polling at a defined interval, such as every 15 seconds, or via subscription-based updates where supported by the third-party media streaming service 132. The polling interval and rate limiting parameters may be adjusted dynamically based on usage requirements and partner constraints.

If the access token has expired or been invalidated since the last retrieval, the social interaction platform 510 may first execute a token refresh operation to obtain a replacement access token before transmitting the retrieval request. If the token refresh operation fails, the social interaction platform 510 may transmit a notification to the first user indicating that the connection to the third-party media streaming service 132 is no longer active.

At operation 1310, the system converts the retrieved data to a unified schema. This operation corresponds to the conversion performed by the normalization module 512 described with reference to FIG. 5 and to the population of the unified media activity structure 604 described with reference to FIG. 6. A partner-specific interface module associated with the third-party media streaming service 132 from which the data was retrieved parses the partner-specific response and maps each partner-specific data field to a corresponding field of the unified schema. The unified schema comprises fields including a user identifier, a track name and artist, a playback state and progress, an event timestamp, a partner service identifier, and an ISRC unique media code, as shown in the unified media activity structure 604 in FIG. 6. For example, the partner track identifier "3WOhcATHxK1X" may be mapped to the track identifier field, the artist field "Peggy Gou" may be mapped to the artist identifier field, the playback state field "playing" may be mapped to the playback state field, and the timestamp "2026-03-12T14:32:07Z" may be mapped to the event timestamp field. The partner service identifier field is populated with a value identifying the third-party media streaming service 132 from which the data originated, such as "streamingservice_1." The ISRC unique media code field is populated with the international standard recording code associated with the track, such as "NLZ542200042," which enables cross-platform track identification.

The conversion operation enables downstream components of the social interaction platform 510 to process the media consumption activity data without dependency on the data format or interface conventions of any particular third-party media streaming service 132. As described with reference to FIG. 5 and claim 8, the same conversion operation is performed for media consumption activity data retrieved from a second third-party media streaming service that is different from the first, with a different partner-specific interface module performing the parsing and mapping for the second service. The resulting standardized activity records from both services conform to the same unified schema and are processed by the same downstream presence logic and rendering components.

At operation 1312, the system determines a presence state using presence state logic. This operation corresponds to the processing performed by the presence logic module 514 described with reference to FIG. 5. The presence logic module 514 evaluates the standardized activity record generated at operation 1310 to determine whether and how the first user's listening activity should be displayed to other users. The presence state determination involves comparing the event timestamp of the standardized activity record against a current time to compute an elapsed duration, and evaluating the playback state field to determine whether the first user is actively playing a track.

The process then proceeds to a decision point at operation 1314, which evaluates whether the elapsed duration exceeds a suppression threshold. The suppression threshold may correspond to the 24-hour inactivity threshold described with reference to the application activity monitor 710 in FIG. 7, beyond which the social interaction platform 510 ceases retrieval of media consumption activity data entirely.

If the elapsed duration exceeds the suppression threshold (the "YES" path from operation 1314), the process proceeds to operation 1316, where the system suppresses the listening status indicator. At operation 1316, the UI update module 516 does not render a listening status indicator for the first user on any surface of the social interaction platform 510. The first user's avatar on the map interface is rendered without a callout element displaying track information. The tray element associated with the first user, if expanded by a viewing user, does not include a listening status indicator or an add song button associated with a currently playing or recently played track. The suppression at operation 1316 may also be triggered by conditions other than elapsed duration, such as the first user having activated the listen privately option 1206 described with reference to FIG. 12, or the viewing user not satisfying the visibility rules specified by the first user's sharing preference selection as configured through the share permission dialog 1106 described with reference to FIG. 11. When the listening status indicator is suppressed, the process terminates for the current evaluation cycle and may be re-initiated at the next polling interval or upon a subsequent triggering event.

If the elapsed duration does not exceed the suppression threshold (the "NO" path from operation 1314), the process proceeds to operation 1318, where the system updates the UI text by removing the word "currently" or adjusting the label to reflect the recency of the listening activity. This operation implements the distinction between a live listening state and a recently listened state. When the elapsed duration is below a first threshold and the playback state indicates active playback, the presence state is classified as a live listening state, and the listening status indicator may display text such as "Listening on Spotify" with language indicating present-tense activity. When the elapsed duration is above the first threshold but below the suppression threshold, the presence state is classified as a recently listened state, and the UI text is updated to reflect past-tense activity, such as changing "Listening on Spotify" to "Listened on Spotify" or removing the word "currently" from any descriptive label. This text update at operation 1318 provides the viewing user with an indication of whether the first user is currently engaged in media playback or whether the displayed track represents a recent but no longer active playback session.

At operation 1320, the system displays the listening status indicator on a social surface. This operation corresponds to the rendering performed by the UI update module 516 described with reference to FIG. 5 and to the causing display operation recited in claim 1. The UI update module 516 renders the listening status indicator on one or more surfaces of the social interaction platform 510 presented on a device of a second user. As described with reference to claim 15 and FIG. 8, the listening status indicator may be rendered on multiple surfaces including a map interface as a callout element positioned proximate to an avatar of the first user, a profile interface as a persistent indicator element, a contextual tray element that is displayed in response to a selection interaction by the second user on the map interface, or a content page associated with a media track identified in the standardized activity record.

The specific visual layout and interactive elements rendered at operation 1320 depend on various criteria. In a first treatment, the add song button 714 may be rendered as a primary call-to-action element within the tray element 712. In a second treatment, the add song button 714 may be rendered alongside a chat element as co-equal interactive elements. In a third treatment, an add element may be rendered within the callout element on the map interface. The UI update module 516 also determines whether to render a platform-specific icon or a generic media icon within the listening status indicator based on whether the viewing user corresponds to the first user or to a different user.

Upon rendering the listening status indicator at operation 1320, the user action logger 716 records an impression event indicating that the listening status indicator was displayed to the viewing user, along with metadata including the surface on which it was rendered, the treatment to which the viewing user was assigned, and the presence state (live listening or recently listened) at the time of rendering. These logged events enable the social interaction platform 510 to compute metrics such as listening status impressions per treatment, tray open rates, and downstream engagement rates.

The flowchart 1304 may also incorporate an operation for handling error states encountered during the retrieval and rendering process. At any point during operations 1308 through 1320, the social interaction platform 510 may encounter an error condition. Error conditions may include a failure to retrieve media consumption activity data from the third-party media streaming service 132 due to a network timeout, an HTTP error response, or a GRPC timeout; a failure of the token refresh operation resulting in an invalidated or expired access token; a failure to match the ISRC or other track identifier to a corresponding entry in the internal media library of the social interaction platform 510; or a failure of a write request transmitted to a streaming service linked to the second user's account when the second user selects the add song button 714. When the social interaction platform 510 encounters an error during the retrieval operation at operation 1308, the system may suppress the listening status indicator for the current evaluation cycle and retry the retrieval at the next polling interval. When the system encounters a token invalidation error, the system may transmit a notification to the first user indicating that the connection to the third-party media streaming service 132 is no longer active, such as a toast banner or modal reading "Unable to link to your Spotify account," and may update the connected streaming service indicator 1208 in the settings interface described with reference to FIG. 12 to reflect the disconnected state.

When the second user selects the add song button 714 and the write request to the second user's linked streaming service fails, the social interaction platform 510 may display an error notification on the device of the second user, such as a toast banner reading "Unable to add song. Try again later." The add song button 714 may remain in the first state indicating an available save action, rather than transitioning to the second state, so that the second user may re-attempt the save operation. The social interaction platform 510 may implement a retry mechanism that automatically retransmits the write request a defined number of times at defined intervals before displaying the error notification to the second user. The user action logger 716 may record the failed save attempt, including the error type and the surface from which the attempt was initiated, to enable the social interaction platform 510 to monitor error rates and identify systemic issues with particular streaming service integrations.

The flowchart 1304 shown in FIG. 13 represents a single evaluation cycle for a single first user. The social interaction platform 510 may execute this process concurrently for multiple first users whose listening activity is to be surfaced to one or more viewing users. The process repeats at each polling interval or upon each triggering event, with the presence state being re-evaluated at each cycle based on the most recent standardized activity record. If the first user's playback state changes between cycles, for example from "playing" to "paused" or from one track to a different track, the updated data is retrieved at operation 1308, converted at operation 1310, evaluated at operation 1312, and the listening status indicator is updated accordingly at operation 1318 and operation 1320.

FIG. 14 is a flowchart illustrating routine 1400 (e.g., a method or process), according to some examples. Although the example method depicted in FIG. 14 depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the method. In some examples, different components of an example device or system that implements the method may perform functions at substantially the same time or in a specific sequence.

In operation 1412, the system (e.g., the interaction client 104) retrieves, from a third-party media streaming service 132 via an application programming interface 508, media consumption activity data associated with a first user of a social interaction platform 510. The retrieval at operation 1412 corresponds to the data flow from the external media service 506 through the API 508 as described with reference to FIG. 5, and to the retrieval operation at operation 1308 in the flowchart 1304 described with reference to FIG. 13. The system transmits a request to the application programming interface of the third-party media streaming service 132 that is linked to the first user's account on the social interaction platform 510. The request includes authentication credentials, such as an access token that was obtained through an OAuth-based authorization flow and stored in association with the first user's account in the profile data 302 within the database 128 as described with reference to FIG. 3. The third-party media streaming service 132 returns a response containing media consumption activity data in a partner-specific format. The media consumption activity data may include fields such as a partner track identifier, an artist name, an album name, a playback state indicating whether the first user is actively playing a media track or has recently played a media track, and a timestamp indicating when the playback event occurred or was last updated.

Prior to the retrieval at operation 1412, the system may verify that the first user has linked a streaming service account to the social interaction platform 510 and that the stored access token is valid. If the access token has expired, the system may execute a token refresh operation to obtain a replacement access token. If the token refresh operation fails, the system may transmit a notification to the first user indicating that the connection to the third-party media streaming service 132 is no longer active, such as a notification reading "Unable to link to your streaming account."

The system may also verify, prior to the retrieval at operation 1412, that the first user satisfies one or more conditions for retrieval to proceed. As described with reference to the application activity monitor 710 in FIG. 7, the system monitors an application activity signal associated with the first user on the social interaction platform 510. If the duration since the last time the first user opened or interacted with the social interaction platform 510 exceeds an inactivity threshold, such as 24 hours, the system ceases retrieval of media consumption activity data from the third-party media streaming service 132 for the first user. In this case, operation 1412 is not performed until the first user reopens the social interaction platform 510, at which point the system resumes retrieval and restores the first user's prior sharing configuration. Additionally, as described with reference to the listen privately option 1206 in FIG. 12, if the first user has activated a private listening mode, the system ceases retrieval of media consumption activity data for the duration specified by the duration parameter selected by the first user. In this case, operation 1412 is not performed until the private listening mode is deactivated, either by expiration of the duration parameter or by manual deactivation by the first user.

The retrieval at operation 1412 may occur via polling at a defined interval or via subscription-based updates where supported by the third-party media streaming service 132. The polling interval may be, for example, 15 seconds. The polling interval and rate limiting parameters may be adjusted dynamically based on usage requirements and constraints imposed by the third-party media streaming service 132.

In operation 1414, the system converts the media consumption activity data into a standardized activity record according to a unified schema. The conversion at operation 1414 corresponds to the processing performed by the normalization module 512 described with reference to FIG. 5 and to the population of the unified media activity structure 604 described with reference to FIG. 6. A partner-specific interface module associated with the third-party media streaming service 132 from which the media consumption activity data was retrieved parses the partner-specific response received at operation 1412 and maps each partner-specific data field to a corresponding field of the unified schema.

The unified schema comprises fields including a user identifier, a track name and artist, a playback state and progress, an event timestamp, a partner service identifier, and an ISRC unique media code, as shown in the unified media activity structure 604 in FIG. 6. The partner-specific interface module extracts one or more partner-specific data fields from the response, maps each extracted field to a corresponding field of the unified schema, and populates the standardized activity record with values derived from the mapping. For example, a partner track identifier "3WOhcATHxK1X" may be mapped to the track identifier field, an artist field "Peggy Gou" may be mapped to the artist identifier field, a playback state field "playing" may be mapped to the playback state field, and a timestamp "2026-03-12T14:32:07Z" may be mapped to the event timestamp field. The partner service identifier field is populated with a value identifying the third-party media streaming service 132 from which the data originated, such as "streamingservice_1." The ISRC unique media code field is populated with the international standard recording code associated with the track, such as "NLZ542200042."

The conversion at operation 1414 enables downstream operations in the routine 1400 to process the media consumption activity data without dependency on the data format or interface conventions of any particular third-party media streaming service 132. When the social interaction platform 510 integrates with multiple third-party media streaming services, each service's media consumption activity data is converted into a standardized activity record according to the same unified schema using a respective partner-specific interface module. The resulting standardized activity records from different services conform to the same schema and are processed by the same downstream presence state determination and rendering operations at operations 1416 and 1426.

In operation 1416, the system determines, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service 132. The determination at operation 1416 corresponds to the processing performed by the presence logic module 514 described with reference to FIG. 5 and to the presence state determination at operation 1312 in the flowchart 1304 described with reference to FIG. 13.

The presence state determination at operation 1416 comprises comparing the event timestamp of the standardized activity record against a current time to compute an elapsed duration. The system then classifies the presence state based on the elapsed duration and the playback state. In response to determining that the elapsed duration is below a first threshold and the playback state indicates active playback, the system classifies the presence state as a live listening state. In response to determining that the elapsed duration is above the first threshold and below a second threshold, the system classifies the presence state as a recently listened state. In response to determining that the elapsed duration exceeds the second threshold, the system classifies the presence state as an inactive state and suppresses display of the listening status indicator for the first user.

The first threshold and the second threshold are configurable values that define the boundaries between presence states. For example, a first threshold of 60 seconds may be used to distinguish between a live listening state and a recently listened state. The second threshold defines the boundary beyond which the system treats the first user's listening activity as stale and suppresses display of the listening status indicator entirely. The thresholds may be configured globally, per partner, or per user, and may be adjusted based on the characteristics of the third-party media streaming service 132. For example, a streaming service that provides true real-time playback signals may use a shorter first threshold than a streaming service that provides only recently-played data.

The presence state determination at operation 1416 also incorporates an evaluation of the first user's sharing preferences and the viewing user's eligibility. As described with reference to the share permission dialog 1106 in FIG. 11, the system evaluates whether a second user satisfies the one or more visibility rules specified by the first user's sharing preference selection. The sharing preference selection may specify one or more visibility rules including sharing with all friends of the first user, sharing with a subset of friends with whom the first user shares location data, or sharing with a custom list of one or more individually selected friends. If the second user does not satisfy the applicable visibility rule, the presence state is treated as suppressed with respect to that second user, and the listening status indicator is not rendered for that second user even if the first user's playback activity would otherwise qualify for a live listening or recently listened state.

The presence state determination at operation 1416 further accounts for whether the first user has activated a private listening mode. As described with reference to the listen privately option 1206 in FIG. 12, if the first user has activated the private listening mode, the system suppresses display of the listening status indicator for the first user to all friends regardless of the sharing preference selection. In this case, the presence state is classified as suppressed for all viewing users.

In operation 1426, the system causes display, within a surface of the social interaction platform 510 presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record. The display operation at operation 1426 corresponds to the rendering performed by the UI update module 516 described with reference to FIG. 5 and to the display operation at operation 1320 in the flowchart 1304 described with reference to FIG. 13.

The listening status indicator may be rendered on one or more surfaces of the social interaction platform 510. The surfaces on which the listening status indicator may be rendered include a map interface as a callout element positioned proximate to an avatar of the first user, a profile interface of the first user as a persistent indicator element displaying at least a track name and an artist name from the standardized activity record, a contextual tray element that is displayed in response to a selection interaction by the second user on the map interface, and a content page associated with a media track identified in the standardized activity record. These surfaces correspond to the map interface 806, the listening status indicator 808, and the privacy settings 810 described with reference to FIG. 8, and to the first user interface 908 and second user interface 910 described with reference to FIG. 9.

When the listening status indicator is rendered on the map interface, the system determines a geographic location of the first user based on one or more location signals received from a device of the first user and renders an avatar representing the first user at a position on the map interface corresponding to the geographic location, as described with reference to FIG. 9. A callout element is generated and positioned proximate to the avatar, the callout element comprising a track name and an artist name extracted from the standardized activity record. The callout element also includes a selectable element that, upon selection by the second user, initiates navigation to a content page associated with the track identified in the standardized activity record.

The system determines whether the viewing user to whom the map interface is being presented corresponds to the first user or to a user other than the first user. In response to determining that the viewing user corresponds to the first user, the system renders a platform-specific icon identifying the third-party media streaming service 132 from which the media consumption activity data was retrieved within the callout element or the listening status indicator. In response to determining that the viewing user corresponds to a user other than the first user, the system renders a generic media icon in place of the platform-specific icon, the generic media icon being visually distinct from the platform-specific icon and not identifying a particular third-party media streaming service. This distinction prevents the map interface from displaying multiple partner-specific logos across the map when a viewing user has multiple friends listening on different streaming services.

When the second user selects the avatar or the callout element on the map interface, the system expands a tray element within the map interface, as described with reference to FIG. 9. The tray element displays the first user's profile identifier, a last-active timestamp associated with the first user, and the listening status indicator. The tray element also renders one or more interactive elements including an add song button 714 configured to initiate a save operation for the track identified in the standardized activity record, a streaming service identifier indicating the third-party media streaming service 132 from which the media consumption activity data was retrieved, a camera icon, and a chat icon. The specific visual layout and interactive elements rendered at operation 1426 may depend on an experimental cohort to which the viewing user is assigned. In a first treatment, the add song button 714 may be rendered as a primary call-to-action element within the tray element 712. In a second treatment, the add song button 714 may be rendered alongside a chat element as co-equal interactive elements. In a third treatment, an add element may be rendered within the callout element on the map interface rather than within the tray element.

When the second user selects the add song button 714 within the tray element, the system performs a save operation. The system identifies a track identifier from the standardized activity record and transmits a write request via an application programming interface associated with a streaming service linked to the second user's account to save the track to the second user's library. If the streaming service linked to the second user's account is different from the third-party media streaming service 132 associated with the first user, the system extracts an international standard recording code from the standardized activity record, matches the code to a corresponding track entry in an internal media library of the social interaction platform 510, resolves a platform-specific track identifier for the second user's streaming service from the corresponding track entry, and transmits the write request using the resolved platform-specific track identifier. Upon receiving a confirmation that the write request was processed, the system updates the add song button 714 from a first state displaying "Add" to a second state displaying "Added" and causes display of a confirmation notification on the device of the second user, such as the notification banner reading "Added to your Liked Songs playlist" shown in FIG. 9 and FIG. 11.

If the second user has not linked a streaming service account to the social interaction platform 510, and the second user selects the add song button 714 or another interactive element associated with the listening status indicator, the system presents a connection prompt to the second user as described with reference to FIG. 10. The connection prompt comprises one or more selectable options each corresponding to a different third-party media streaming service 132 available for linking, such as the first streaming service button and the second streaming service button shown in FIG. 10. The system initiates an authorization flow with the streaming service selected by the second user and logs the interaction and the surface from which the connection prompt was presented as a conversion event. The system records an entry point identifier for each presentation of the connection prompt, computes conversion rates for each entry point, and may adjust the frequency or priority of presenting the connection prompt at each entry point based on the computed conversion rates. Entry points from which the connection prompt may be presented include a map tray element, a content page associated with a media track, a settings interface such as the user interface 1204 described with reference to FIG. 12, a profile indicator element, and a feed banner element.

The system may also render a deep link within the listening status indicator or within the tray element. When the second user selects the listening status indicator or a portion thereof, the system generates a deep link comprising one or more parameters identifying the track indicated in the standardized activity record. The system determines whether a native application associated with a target streaming service is installed on the device of the second user. If the native application is installed, the system transmits the deep link to the native application to cause it to navigate to a content view associated with the track. If the native application is not installed, the system redirects the second user to a web-based interface or an application store listing associated with the target streaming service.

The user action logger 716 described with reference to FIG. 7 records events associated with each operation in the routine 1400. These events include retrieval events at operation 1412, conversion events at operation 1414, presence state classification events at operation 1416, and display events and user interaction events at operation 1426. The recorded events enable the social interaction platform 510 to compute metrics including connection rates by surface, tray open rates, add song tap rates, deep link tap-through rates, and listening status impressions per experimental treatment. These metrics inform the optimization of the routine 1400 and the associated user interface treatments across the social interaction platform 510.

The social interaction platform 510 may also render a listening status indicator within a chat interface of the social interaction platform 510. When a first user who has linked a streaming service account and is actively consuming media opens a chat conversation with a second user, the social interaction platform 510 may render a chat header element within the chat interface. The chat header element may display the first user's display name, an indication that the first user is currently listening to media, and a track name and artist name extracted from the standardized activity record associated with the first user. The chat header element may be rendered above the message history within the chat interface and may be visible to the second user when the second user opens the same chat conversation. The chat header element provides an additional surface through which the second user may discover the first user's listening activity without navigating to the map interface or the first user's profile. The chat header element may include a selectable element that, upon selection by the second user, initiates navigation to a content page associated with the track identified in the standardized activity record, or expands to display an add song button configured to initiate a save operation for the track.

The social interaction platform 510 may also render a listening status indicator within a Friends Feed surface. The Friends Feed surface presents a list of recent interactions, messages, and status updates associated with friends within the user's social graph. When a first user is actively consuming media on a linked third-party media streaming service 132, the social interaction platform 510 may display a banner element or an inline status element within the Friends Feed surface indicating that the first user is listening to a particular track. The banner element may include the first user's display name, a track name and artist name, and a streaming service indicator. The banner element may be selectable, and upon selection, may navigate the second user to the map interface centered on the first user's location, or may expand to display additional information and interactive elements such as an add song button. The Friends Feed surface serves as one of the multiple entry points from which the social interaction platform 510 may present a connection prompt to a user who has not yet linked a streaming service account, and the user action logger 716 may record the Friends Feed banner as an entry point identifier when logging conversion events.

The social interaction platform 510 may further render a listening status indicator within a profile interface associated with the first user. The profile interface may be the first user's own profile as viewed by the first user, or a friendship profile as viewed by a second user. When the first user is actively consuming media, the social interaction platform 510 may render a persistent indicator element, such as a pill or badge, within the profile interface. The persistent indicator element may display a track name and artist name extracted from the standardized activity record, and may include a streaming service icon. In some examples, the persistent indicator element is rendered adjacent to or below the first user's display name, avatar, or Bitmoji representation on the profile interface. When the first user is not actively consuming media but has a recently listened standardized activity record, the persistent indicator element may display the most recently played track with an indication that the track was played at a prior time. Selection of the persistent indicator element by a viewing user may navigate the viewing user to a content page associated with the track, or may present a connection prompt if the viewing user has not yet linked a streaming service account. The persistent indicator element on the profile interface also serves as a connection entry point; when the first user views the first user's own profile and has not yet linked a streaming service account, the social interaction platform 510 may render the persistent indicator element as a prompt to connect a streaming service, such as displaying text reading "Start Listening on Spotify" or a similar call-to-action label that, upon selection, initiates the account linking flow described with reference to FIG. 10.

SYSTEM WITH HEAD-WEARABLE APPARATUS

FIG. 15 illustrates a system 1500 including a head-wearable apparatus 116 with a selector input device, according to some examples. FIG. 15 is a high-level functional block diagram of an example head-wearable apparatus 116 communicatively coupled to a mobile device 114 and various server systems 1504 (e.g., the server system 110) via various networks.

The head-wearable apparatus 116 includes one or more cameras, each of which may be, for example, a visible light camera 1506, an infrared emitter 1508, and an infrared camera 1510.

The mobile device 114 connects with head-wearable apparatus 116 using both a low-power wireless connection 1512 and a high-speed wireless connection 1514. The mobile device 114 is also connected to the server system 1504 and the network 1316.

The head-wearable apparatus 116 further includes two image displays of the image display of optical assembly 1518. The two image displays of optical assembly 1518 include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus 116. The head-wearable apparatus 116 also includes an image display driver 1520, an image processor 1522, low-power circuitry 1524, and high-speed circuitry 1526. The image display of optical assembly 1518 is for presenting images and videos, including an image that can include a graphical user interface to a user of the head-wearable apparatus 116.

The image display driver 1520 commands and controls the image display of optical assembly 1518. The image display driver 1520 may deliver image data directly to the image display of optical assembly 1518 for presentation or may convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H.264 (MPEG-4 Part 10), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to compression formats such as Portable Network Graphics (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (EXIF) or the like.

The head-wearable apparatus 116 includes a frame and stems (or temples) extending from a lateral side of the frame. The head-wearable apparatus 116 further includes a user input device 1528 (e.g., touch sensor or push button), including an input surface on the head-wearable apparatus 116. The user input device 1528 (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

The components shown in FIG. 15 for the head-wearable apparatus 116 are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively, or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus 116. Left and right visible light cameras 1506 can include digital camera elements such as a complementary metal oxide–semiconductor (CMOS) image sensor, charge-coupled device, camera lenses, or any other respective visible or light-capturing elements that may be used to capture data, including images of scenes with unknown objects.

The head-wearable apparatus 116 includes a memory 1502, which stores instructions to perform a subset, or all the functions described herein. The memory 1502 can also include a storage device.

As shown in FIG. 15, the high-speed circuitry 1526 includes a high-speed processor 1530, a memory 1502, and high-speed wireless circuitry 1532. In some examples, the image display driver 1520 is coupled to the high-speed circuitry 1526 and operated by the high-speed processor 1530 to drive the left and right image displays of the image display of optical assembly 1518. The high-speed processor 1530 may be any processor capable of managing high-speed communications and operation of any general computing system needed for the head-wearable apparatus 116. The high-speed processor 1530 includes processing resources needed for managing high-speed data transfers on a high-speed wireless connection 1514 to a wireless local area network (WLAN) using the high-speed wireless circuitry 1532. In certain examples, the high-speed processor 1530 executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus 116, and the operating system is stored in the memory 1502 for execution. In addition to any other responsibilities, the high-speed processor 1530 executing a software architecture for the head-wearable apparatus 116 is used to manage data transfers with high-speed wireless circuitry 1532. In certain examples, the high-speed wireless circuitry 1532 is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as WI-FI®. In some examples, other high-speed communications standards may be implemented by the high-speed wireless circuitry 1532.

The low-power wireless circuitry 1534 and the high-speed wireless circuitry 1532 of the head-wearable apparatus 116 can include short-range transceivers (e.g., Bluetooth™ , Bluetooth LE, Zigbee, ANT+) and wireless wide, local, or wide area network transceivers (e.g., cellular or WI-FI®). Mobile device 114, including the transceivers communicating via the low-power wireless connection 1512 and the high-speed wireless connection 1514, may be implemented using details of the architecture of the head-wearable apparatus 116, as can other elements of the network 1316.

The memory 1502 includes any storage device capable of storing various data and applications, including, among other things, camera data generated by the left and right visible light cameras 1506, the infrared camera 1510, and the image processor 1522, as well as images generated for display by the image display driver 1520 on the image displays of the image display of optical assembly 1518. While the memory 1502 is shown as integrated with high-speed circuitry 1526, in some examples, the memory 1502 may be an independent standalone element of the head-wearable apparatus 116. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor 1530 from the image processor 1522 or the low-power processor 1536 to the memory 1502. In some examples, the high-speed processor 1530 may manage addressing of the memory 1502 such that the low-power processor 1336 will boot the high-speed processor 1530 any time that a read or write operation involving memory 1502 is needed.

As shown in FIG. 15, the low-power processor 1536 or high-speed processor 1530 of the head-wearable apparatus 116 can be coupled to the camera (visible light camera 1506, infrared emitter 1508, or infrared camera 1510), the image display driver 1520, the user input device 1528 (e.g., touch sensor or push button), and the memory 1502.

The head-wearable apparatus 116 is connected to a host computer. For example, the head-wearable apparatus 116 is paired with the mobile device 114 via the high-speed wireless connection 1514 or connected to the server system 1504 via the network 1316. The server system 1504 may be one or

more computing devices as part of a service or network computing system, for example, that includes a processor, a memory, and network communication interface to communicate over the network 1316 with the mobile device 114 and the head-wearable apparatus 116.
The mobile device 114 includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network 1316, low-power wireless connection 1512, or high-speed wireless connection 1514. The mobile device 114 can further store at least portions of the instructions in the memory of the mobile device 114 memory to implement the functionality described herein.
Output components of the head-wearable apparatus 116 include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light-emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver 1520. The output components of the head-wearable apparatus 116 further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus 116, the mobile device 114, and server system 1504, such as the user input device 1528, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.
The head-wearable apparatus 116 may also include additional peripheral device elements. Such peripheral device elements may include sensors and display elements integrated with the head-wearable apparatus 116. For example, peripheral device elements may include any input/output (I/O) components including output components, motion components, position components, or any other such elements described herein.
In some examples, the head-wearable apparatus 116 may include biometric components or sensors s to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The biometric components may include a brain-machine interface (BMI) system that allows communication between the brain and an external device or machine. This may be achieved by recording brain activity data, translating this data into a format that can be understood by a computer, and then using the resulting signals to control the device or machine.
Example types of BMI technologies, include:

- Electroencephalography (EEG) based BMIs, which record electrical activity in the brain using electrodes placed on the scalp.
- Invasive BMIs, which used electrodes that are surgically implanted into the brain.
- Optogenetics BMIs, which use light to control the activity of specific nerve cells in the brain.

Any biometric data collected by the biometric components is captured and stored with only user approval and deleted on user request, and in accordance with applicable laws. Further, such biometric data may be used for very limited purposes, such as identification verification. To ensure limited and authorized use of biometric information and other personally identifiable information (PII), access to this data is restricted to authorized personnel only, if at all. Any use of biometric data may strictly be limited to identification verification purposes, and the biometric data is not shared or sold to any third party without the explicit consent of the user. In addition, appropriate technical and organizational measures are implemented to ensure the security and confidentiality of this sensitive information.
The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), Wi-Fi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over low-power wireless connections 1512 and high-speed wireless connection 1514 from the mobile device 114 via the low-power wireless circuitry 1534 or high-speed wireless circuitry 1532.
MACHINE ARCHITECTURE
FIG. 16 is a diagrammatic representation of a machine 1600 within which instructions 1602 (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine 1600 to perform any one or more of the methodologies discussed herein.. The instructions 1602 transform the general, non-programmed machine 1600 into a particular machine 1600 programmed to carry out the described and illustrated functions in the manner described. The machine 1600 may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine 1600 may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine 1600 may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions 1602, sequentially or otherwise, that specify actions to be taken by the machine 1600. Further, while a single machine 1600 is illustrated, the term "machine" shall also be taken to include a collection of machines that individually or jointly execute the instructions 1602 to perform any one or more of the methodologies discussed herein. The machine 1600, for example, may comprise the user system 102 or any one of multiple server devices forming part of the server system 110. In some examples, the machine 1600 may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the method or algorithm being performed on the client-side.
The machine 1600 may include processors 1604, memory 1606, and input/output I/O components 1608, which may be configured to communicate with each other via a bus 1610.
The memory 1606 includes a main memory 1616, a static memory 1618, and a storage unit 1620, both accessible to the processors 1604 (e.g., processor 1612 or processor 1614 via the bus 1610. The main memory 1606, the static memory 1618, and storage unit 1620 store the instructions 1602 embodying any one or more of the methodologies or functions described herein. The instructions 1602 may also reside, completely or partially, within the main memory 1616, within the static memory 1618, within machine-readable medium 1622 within the storage unit 1620, within at least one of the processors 1604 (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine 1600.
The I/O components 1608 may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components 1608 that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components 1608 may include many other

components that are not shown in FIG. 16. In various examples, the I/O components 1608 may include user output components 1624 and user input components 1626. The user output components 1624 may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components 1626 may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further examples, the I/O components 1608 may include biometric components 1628, motion components 1630, environmental components 1632, or position components 1634, among a wide array of other components. For example, the biometric components 1628 include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like.

The motion components 1630 include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

The environmental components 1632 include, for example, one or more cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detect concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

With respect to cameras, the user system 102 may have a camera system comprising, for example, front cameras on a front surface of the user system 102 and rear cameras on a rear surface of the user system 102. The front cameras may, for example, be used to capture still images and video of a user of the user system 102 (e.g., "selfies"), which may then be modified with digital effect data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being modified with digital effect data. In addition to front and rear cameras, the user system 102 may also include a 360° camera for capturing 360° photographs and videos.

Moreover, the camera system of the user system 102 may be equipped with advanced multi-camera configurations. This may include dual rear cameras, which might consist of a primary camera for general photography and a depth-sensing camera for capturing detailed depth information in a scene. This depth information can be used for various purposes, such as creating a bokeh effect in portrait mode, where the subject is in sharp focus while the background is blurred. In addition to dual camera setups, the user system 102 may also feature triple, quad, or even penta camera configurations on both the front and rear sides of the user system 102. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

Communication may be implemented using a wide variety of technologies. The I/O components 1608 further include communication components 1436 operable to couple the machine 1600 to a network 1638 or devices 1640 via respective couplings or connections. For example, the communication components 1636 may include a network interface component or another suitable device to interface with the network 1638. In further examples, the communication components 1636 may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices 1640 may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components 1636 may detect identifiers or include components operable to detect identifiers. For example, the communication components 1636 may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar codes, multi-dimensional bar codes such as Quick Response (QR) codes, Aztec codes, Data Matrix, Dataglyph™, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar codes, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components 1636, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., main memory 1616, static memory 1618, and memory of the processors 1604) and storage unit 1620 may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions 1602), when executed by processors 1604, cause various operations to implement the disclosed examples.

The instructions 1602 may be transmitted or received over the network 1638, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components 1636) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol [HTTP]). Similarly, the instructions 1602 may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices 1640.

SOFTWARE ARCHITECTURE

FIG. 17 is a block diagram 1700 illustrating a software architecture 1702, which can be installed on any one or more of the devices described herein. The software architecture 1702 is supported by hardware such as a machine 1704 that includes processors 1706, memory 1708, and I/O components 1710. In this example, the software architecture 1702 can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture 1702 includes layers such as an operating system 1712, libraries 1714, frameworks 1716, and applications 1718. Operationally, the applications 1718 invoke API calls 1720 through the software stack and receive messages 1722 in response to the API calls 1720.

The operating system 1712 manages hardware resources and provides common services. The operating system 1712 includes, for example, a kernel 1724, services 1726, and drivers 1728. The kernel 1724 acts as an abstraction layer between the hardware and the other software layers. For example, the kernel 1724 provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services 1726 can provide other common services for the other software layers. The drivers 1728 are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers 1728 can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

The libraries 1714 provide a common low-level infrastructure used by the applications 1718. The libraries 1714 can include system libraries 1730 (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries 1714 can include API libraries 1732 such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries 1714 can also include a wide variety of other libraries 1734 to provide many other APIs to the applications 1718.

The frameworks 1716 provide a common high-level infrastructure that is used by the applications 1718. For example, the frameworks 1716 provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks 1716 can provide a broad spectrum of other APIs that can be used by the applications 1718, some of which may be specific to a particular operating system or platform.

In an example, the applications 1718 may include a home application 1736, a contacts application 1738, a browser application 1740, a book reader application 1742, a location application 1744, a media application 1746, a messaging application 1748, a game application 1750, and a broad assortment of other applications such as a third-party application 1752. The applications 1718 are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications 1718, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application 1752 (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of a platform) may be mobile software running on a mobile operating system such as IOS™ , ANDROID™ , WINDOWS® Phone, or another mobile operating system. In this example, the third-party application 1752 can invoke the API calls 1720 provided by the operating system 1712 to facilitate functionalities described herein.

As used in this disclosure, phrases of the form “at least one of an A, a B, or a C,” “at least one of A, B, or C,” “at least one of A, B, and C,” and the like, should be interpreted to select at least one from the group that comprises “A, B, and C.” Unless explicitly stated otherwise in connection with a particular instance in this disclosure, this manner of phrasing does not mean “at least one of A, at least one of B, and at least one of C.” As used in this disclosure, the example “at least one of an A, a B, or a C,” would cover any of the following selections: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, and {A, B, C}.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense, e.g., in the sense of “including, but not limited to.”

As used herein, the terms “connected,” “coupled,” or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof.

Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, refer to this application as a whole and not to any portions of this application. Where the context permits, words using the singular or plural number may also include the plural or singular number respectively.

The word “or” in reference to a list of two or more items, covers all the following interpretations of the word: any one of the items in the list, all the items in the list, and any combination of the items in the list. Likewise, the term “and/or” in reference to a list of two or more items, covers all the following interpretations of the word: any one of the items in the list, all the items in the list, and any combination of the items in the list.

The various features, operations, or processes described herein may be used independently of one another, or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure. In addition, certain method or process blocks may be omitted in some implementations.

Although some examples, e.g., those depicted in the drawings, include a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the functions as described in the examples. In other examples, different components of an example device or system that implements an example method may perform functions at substantially the same time or in a specific sequence.

As used herein, the term "social interaction platform" refers to a software application or networked system that enables users to establish connections with other users, communicate with those users, and share content or status information within a social graph. A social interaction platform maintains user accounts, friend or follower relationships, and one or more user interfaces through which users interact with the platform and with each other. A social interaction platform may include multiple surfaces, such as without limitation, a map interface displaying geographic locations of users, a profile interface displaying user information and status, a messaging or chat interface, a content feed interface, a settings interface, and one or more content pages. The social interaction platform operates across one or more client devices and one or more backend servers, and may communicate with external services including one or more third-party media streaming services.

As used herein, the term "surface" refers to a distinct area, view, screen, or user interface component within the social interaction platform on which content, status information, or interactive elements may be rendered and presented to a user. A surface may correspond to, without limitation, a map interface that displays geographic locations of users and associated status information, a profile interface that displays information about a particular user, a contextual tray that expands within another surface in response to a user interaction, a content page associated with a media track or audio topic, a settings interface for configuring account and feature preferences, a feed interface displaying updates from friends or contacts, or a banner or promotional element displayed within the social interaction platform. The system may render a listening status indicator on one or more surfaces concurrently, and different surfaces may present the listening status indicator in different visual formats, layouts, or levels of detail. The system may also present onboarding prompts, connection prompts, and other interactive elements on one or more surfaces.

As used herein, the term "third-party media streaming service" refers to an external platform or service, separate from the social interaction platform, that provides on-demand access to media content for playback by users. A third-party media streaming service may provide access to music tracks, podcasts, audiobooks, video content, or other forms of digital media. Examples of a third-party media streaming service include, but are not limited to, music streaming platforms, audio content platforms, and video streaming platforms. A third-party media streaming service exposes one or more application programming interfaces through which external systems may retrieve data regarding user accounts, media catalogs, playback activity, and library management operations. A single social interaction platform may integrate with one or more third-party media streaming services concurrently, and different users of the social interaction platform may be associated with different third-party media streaming services.

As used herein, the term "media consumption activity data" refers to data that describes or characterizes a user's interaction with media content on a third-party media streaming service. Media consumption activity data may include, without limitation, an identifier of a media track being played or recently played, an identifier of the artist or creator associated with the media track, an identifier of an album or collection containing the media track, a playback state indicating whether the media track is being actively played or was recently played, a timestamp indicating when the playback event occurred or was last updated, and a session identifier associated with the playback session. Media consumption activity data may be retrieved from a third-party media streaming service via one or more application programming interface calls, and the format, fields, latency, and granularity of such

data may vary across different third-party media streaming services. Media consumption activity data encompasses both real-time playback signals and recently-played or historical playback records.
As used herein, the term "standardized activity record" refers to a data structure that represents media consumption activity in a uniform format according to a unified schema, regardless of which third-party media streaming service originally provided the underlying data. A standardized activity record is generated by converting media consumption activity data received from a third-party media streaming service into the unified schema. The unified schema may define one or more fields including, without limitation, a user identifier, a track identifier, an artist identifier, an album identifier, a playback state, a timestamp, a session identifier, and a partner identifier indicating the third-party media streaming service from which the data originated. The standardized activity record enables downstream components of the system, such as a presence state engine and a surface rendering engine, to process and render media consumption information without dependency on the data format or interface conventions of any particular third-party media streaming service. A partner-specific interface module may perform the conversion by parsing a response received from the application programming interface of a given third-party media streaming service, mapping partner-specific data fields to corresponding fields of the unified schema, and populating the standardized activity record with the mapped values.
As used herein, the term "playback state" refers to a data field within a standardized activity record that indicates the current or most recent status of a user's media playback on a third-party media streaming service. A playback state may take one of a plurality of values including, without limitation, a value indicating that the user is actively playing a media track, a value indicating that playback is paused, and a value indicating that the media track was recently played but playback has since ceased. The playback state may be derived directly from data provided by the third-party media streaming service, or it may be inferred by the system based on other available data such as a timestamp associated with the most recent playback event. Different third-party media streaming services may expose playback state information with varying levels of granularity and latency; for example, one service may provide a real-time indication of active playback while another service may provide only a record of the most recently played track without an explicit indication of whether playback is ongoing.
As used herein, the term "presence state" refers to a classification determined by the system that characterizes the nature and recency of a user's media consumption activity for the purpose of controlling whether and how a listening status indicator is displayed to other users. The presence state is determined based at least on the playback state and the timestamp contained in a standardized activity record. The presence state may take one of a plurality of values including, without limitation, a live listening state indicating that the user is currently consuming media, a recently listened state indicating that the user consumed media within a defined recency threshold but is not currently consuming media, and an inactive state indicating that the user's media consumption activity is stale or absent beyond a defined threshold. The presence state may also reflect the application of one or more privacy or visibility rules, such that even if a user has recent media consumption activity, the presence state may be set to suppress display of the listening status indicator if the user has activated a private listening mode or if the viewing user does not satisfy the sharing preferences configured by the consuming user. The system may apply one or more configurable thresholds to transition between presence states.
As used herein, the term "listening status indicator" refers to a user interface element rendered within one or more surfaces of the social interaction platform that conveys information about a user's media consumption activity to one or more other users. A listening status indicator comprises at least a portion of a standardized activity record, which may include, without limitation, a track name, an artist name, an album name, a streaming service identifier, and a playback state representation. A listening status indicator may be rendered in various visual forms depending on the surface on which it is displayed and the user interface treatment applied; for example, a listening status indicator may appear as a callout element positioned proximate to a user's avatar on a map interface, as a persistent pill or badge on a profile interface, as a row within an expanded tray element, or as an embedded element within a content page. A listening status indicator may also include one or more interactive elements, such as a save element configured to initiate a save operation for the indicated track, a deep link element configured to redirect a viewer to the corresponding track within a streaming application, or a chat element configured to initiate a communication session with the user whose media consumption activity is being displayed. The visual configuration of a listening status indicator, including the placement, prominence, and behavior of its interactive elements, may vary based on an experimental cohort assignment or a partner-specific display requirement.
EXAMPLE STATEMENTS
Example 1: A system comprising: at least one processor; and at least one memory component storing instructions that, when executed by the at least one processor, cause the at least one processor to perform operations comprising: retrieving, from a third-party media streaming service via an application programming interface, media consumption activity data associated with a first user of a social interaction platform; converting the media consumption activity data into a standardized activity record according to a unified schema; determining, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service; and causing display, within a surface of the social interaction platform presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record.
Example 2: The system of Example 1, wherein the operations further comprise: executing an OAuth-based authorization flow with the third-party media streaming service on behalf of the first user, the OAuth-based authorization flow comprising transmitting an authorization request to the third-party media streaming service and receiving an access token in response to the first user granting consent; storing the access token in association with an account of the first user on the social interaction platform; detecting that the access token has expired or been invalidated by the third-party media streaming service; and executing a token refresh operation to obtain a replacement access token, or, in response to determining that the token refresh operation has failed, transmitting a notification to the first user indicating that a connection to the third-party media streaming service is no longer active.
Example 3: The system of any one of Examples 1-2, wherein the surface comprises a map interface, and wherein causing display of the listening status indicator comprises: determining a geographic location of the first user based on one or more location signals received from a device of the first user; rendering an avatar representing the first user at a position on the map interface corresponding to the geographic location of the first user; generating a callout element positioned proximate to the avatar, the callout element comprising a track name and an artist name extracted from the standardized activity record; and rendering, within the callout element, a selectable element that, upon selection by the second user, initiates navigation to a content page associated with a track identified in the standardized activity record.
Example 4: The system of Example 3, wherein the operations further comprise: detecting a selection by the second user of the avatar or the callout element on the map interface; in response to the detecting the selection, expanding a tray element within the map interface, the tray element comprising a profile identifier of the first user, a last-active timestamp associated with the first user, and the listening status indicator; rendering, within the tray element, one or more interactive elements comprising at least an add element configured to initiate a save operation for a track identified in the standardized activity record; and rendering, within the tray element, a streaming service identifier indicating the third-party media streaming service from which the media consumption activity data was retrieved.
Example 5: The system of any one of Examples 3-4, wherein the operations further comprise: detecting a selection of the add element by the second user; identifying, from the standardized activity record, a track identifier associated with the track; transmitting, via a second application programming

interface associated with a streaming service linked to an account of the second user, a write request to save the track to a library of the second user on the streaming service linked to the account of the second user; and in response to receiving a confirmation that the write request was processed, updating the add element from a first state indicating an available save action to a second state indicating that the track has been saved, and causing display of a confirmation notification on the device of the second user.
Example 6: The system of Example 5, wherein the streaming service linked to the account of the second user is different from the third-party media streaming service associated with the first user, and wherein the operations further comprise: extracting an international standard recording code from the standardized activity record associated with the track; matching the international standard recording code to a corresponding track entry in an internal media library of the social interaction platform; resolving, from the corresponding track entry, a platform-specific track identifier for the streaming service linked to the account of the second user; and transmitting the write request to the streaming service linked to the account of the second user using the platform-specific track identifier resolved from the corresponding track entry.
Example 7: The system of any one of Examples 3-6, wherein the operations further comprise: determining whether a viewing user to whom the map interface is being presented corresponds to the first user or to a user other than the first user; in response to determining that the viewing user corresponds to the first user, rendering, within the callout element or the listening status indicator associated with the first user on the map interface, a platform-specific icon identifying the third-party media streaming service from which the media consumption activity data was retrieved; and in response to determining that the viewing user corresponds to a user other than the first user, rendering, within the callout element or the listening status indicator associated with the first user on the map interface, a generic media icon in place of the platform-specific icon, the generic media icon being visually distinct from the platform-specific icon and not identifying a particular third-party media streaming service.
Example 8: The system of any one of Examples 1-7, wherein the operations further comprise: retrieving, from a second third-party media streaming service different from the third-party media streaming service, second media consumption activity data associated with a third user of the social interaction platform; parsing the second media consumption activity data using a partner-specific interface module associated with the second third-party media streaming service, the partner-specific interface module mapping one or more fields of the second media consumption activity data to corresponding fields of the unified schema; and converting the second media consumption activity data into a second standardized activity record according to the unified schema, the second standardized activity record comprising a partner identifier distinguishing the second third-party media streaming service from the third-party media streaming service, wherein the determining the presence state and the causing display are performed using the unified schema regardless of which third-party media streaming service provided the media consumption activity data.
Example 9: The system of Example 8, wherein the unified schema comprises one or more fields selected from a group consisting of a user identifier, a track identifier, an artist identifier, an album identifier, a playback state, a timestamp, a session identifier, and a partner identifier, and wherein converting the media consumption activity data into the standardized activity record comprises: parsing a response received from the application programming interface of the third-party media streaming service to extract one or more partner-specific data fields; mapping each of the one or more partner-specific data fields to a corresponding field of the unified schema; and populating the standardized activity record with values derived from the mapping.
Example 10: The system of any one of Examples 1-9, wherein determining the presence state comprises: comparing the timestamp of the standardized activity record against a current time to compute an elapsed duration; in response to determining that the elapsed duration is below a first threshold and the playback state indicates active playback, classifying the presence state as a live listening state; in response to determining that the elapsed duration is above the first threshold and below a second threshold, classifying the presence state as a recently listened state; and in response to determining that the elapsed duration exceeds the second threshold, classifying the presence state as an inactive state and suppressing display of the listening status indicator for the first user.
Example 11: The system of Example 10, wherein the operations further comprise: monitoring an application activity signal associated with the first user on the social interaction platform, the application activity signal indicating a last time the first user opened or interacted with the social interaction platform; determining that a duration since the last time the first user opened or interacted with the social interaction platform exceeds an inactivity threshold; in response to determining that the duration exceeds the inactivity threshold, ceasing retrieval of the media consumption activity data from the third-party media streaming service for the first user; and in response to detecting a subsequent opening of the social interaction platform by the first user, resuming retrieval of the media consumption activity data from the third-party media streaming service and restoring a prior sharing configuration associated with the first user.
Example 12: The system of Example 11, wherein the operations further comprise: detecting that the first user has reopened the social interaction platform after the duration exceeded the inactivity threshold; determining that the first user is actively consuming media on the third-party media streaming service at a time of the reopening; generating a reminder notification for display within the social interaction platform on the device of the first user, the reminder notification informing the first user that the listening status indicator is visible to one or more friends of the first user; and presenting, within the reminder notification, one or more selectable options comprising at least an option to continue sharing and an option to activate a private listening mode.
Example 13: The system of any one of Examples 1-12, wherein the operations further comprise: receiving, from the first user, a sharing preference selection, the sharing preference selection specifying one or more visibility rules for the listening status indicator, the one or more visibility rules comprising at least one rule selected from a group consisting of sharing with all friends of the first user, sharing with a subset of friends with whom the first user shares location data, and sharing with a custom list of one or more individually selected friends; storing the sharing preference selection in association with an account of the first user; prior to causing display of the listening status indicator, evaluating whether the second user satisfies the one or more visibility rules specified by the sharing preference selection; and causing display of the listening status indicator to the second user only in response to determining that the second user satisfies the one or more visibility rules.
Example 14: The system of Example 13, wherein the operations further comprise: receiving, from the first user, a selection of a private listening mode, the selection specifying a duration parameter, the duration parameter indicating one of a fixed time period or an indefinite period; in response to receiving the selection of the private listening mode, suppressing display of the listening status indicator for the first user to all friends of the first user regardless of the sharing preference selection; ceasing retrieval of the media consumption activity data from the third-party media streaming service for the first user for the duration specified by the duration parameter; and in response to determining that the duration parameter has elapsed, deactivating the private listening mode and resuming retrieval of the media consumption activity data and display of the listening status indicator according to the sharing preference selection.
Example 15: The system of any one of Examples 1-14, wherein causing display of the listening status indicator comprises rendering the listening status indicator on one or more surfaces of the social interaction platform, and wherein the operations further comprise at least one of: rendering the listening status indicator within a map interface as a callout element positioned proximate to an avatar of the first user; rendering the listening status indicator within a profile interface of the first user as a persistent indicator element displaying at least a track name and an artist name from the standardized activity record; rendering the listening status indicator within a contextual tray element that is displayed in response to a selection

interaction by the second user on the map interface; or rendering the listening status indicator within a content page associated with a media track identified in the standardized activity record.
Example 16: The system of any one of Examples 1-15, wherein the operations further comprise: detecting that the second user has not linked a streaming service account to the social interaction platform; detecting an interaction by the second user with the listening status indicator associated with the first user, the interaction comprising a selection of a save element rendered within the listening status indicator; in response to detecting the interaction, presenting a connection prompt to the second user, the connection prompt comprising one or more selectable options each corresponding to a different third-party media streaming service available for linking; initiating an authorization flow with a third-party media streaming service selected by the second user from the one or more selectable options; and logging the interaction and a surface from which the connection prompt was presented as a conversion event.
Example 17: The system of Example 16, wherein the operations further comprise: presenting the connection prompt from one or more entry points within the social interaction platform, the one or more entry points comprising at least two of a map tray element, a content page associated with a media track, a settings interface, a profile indicator element, or a feed banner element; recording, for each presentation of the connection prompt, an entry point identifier indicating which of the one or more entry points triggered the presentation; computing a conversion rate for each entry point identifier based on a ratio of completed streaming service linkings to presentations of the connection prompt from that entry point; and adjusting a frequency or a priority of presenting the connection prompt at each of the one or more entry points based on the computed conversion rates.
Example 18: The system of any one of Examples 1-17, wherein the operations further comprise: detecting a selection by the second user of the listening status indicator or a portion thereof; generating a deep link comprising one or more parameters identifying a track indicated in the standardized activity record, the one or more parameters comprising at least a partner-specific track identifier compatible with the third-party media streaming service or a streaming service linked to the second user; determining whether a native application associated with a target streaming service is installed on the device of the second user; in response to determining that the native application is installed, transmitting the deep link to the native application to cause the native application to navigate to a content view associated with the track; and in response to determining that the native application is not installed, redirecting the second user to a web-based interface or an application store listing associated with the target streaming service.
Example 19: A computer-implemented method comprising: retrieving, from a third-party media streaming service via an application programming interface, media consumption activity data associated with a first user of a social interaction platform; converting the media consumption activity data into a standardized activity record according to a unified schema; determining, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service; and causing display, within a surface of the social interaction platform presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record.
Example 20: A non-transitory computer-readable storage medium storing instructions that, when executed by at least one processor, cause the at least one processor to perform operations comprising: retrieving, from a third-party media streaming service via an application programming interface, media consumption activity data associated with a first user of a social interaction platform; converting the media consumption activity data into a standardized activity record according to a unified schema; determining, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service; and causing display, within a surface of the social interaction platform presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record.
TERM EXAMPLES
"Carrier signal" may include, for example, any intangible medium that can store, encoding, or carrying instructions for execution by the machine and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.
"Client device" may include, for example, any machine that interfaces to a network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.
"Component" may include, for example, a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A "hardware component" is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations.
A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processors. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase "hardware component"(or "hardware-implemented component") should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time.
For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information

to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein.

As used herein, "processor-implemented component" may refer to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a "cloud computing" environment or as a "software as a service" (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

"Computer-readable storage medium" may include, for example, both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

"Machine storage medium" may include, for example, a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines, and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media, and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Field-Programmable Gate Arrays (FPGA), flash memory devices, Solid State Drives (SSD), and Non-Volatile Memory Express (NVMe) devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM, DVD-ROM, Blu-ray Discs, and Ultra HD Blu-ray discs. In addition, machine storage medium may also refer to cloud storage services, Network Attached Storage (NAS), Storage Area Networks (SAN), and object storage devices. The terms "machine-storage medium," "device-storage medium," and "computer-storage medium" mean the same thing and may be used interchangeably in this disclosure. The terms "machine-storage media," "computer-storage media," and "device-storage media" specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term "signal medium."

"Network" may include, for example, one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a Virtual Private Network (VPN), a Local Area Network (LAN), a Wireless LAN (WLAN), a Wide Area Network (WAN), a Wireless WAN (WWAN), a Metropolitan Area Network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a Voice over IP (VoIP) network, a cellular telephone network, a 5G™ network, a wireless network, a Wi-Fi® network, a Wi-Fi 6® network, a Li-Fi network, a Zigbee® network, a Bluetooth® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as third Generation Partnership Project (3GPP) including 4G, fifth-generation wireless (5G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

"Non-transitory computer-readable storage medium" may include, for example, a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

"Processor" may include, for example, data processors such as a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), a Quantum Processing Unit (QPU), a Tensor Processing Unit (TPU), a Neural Processing Unit (NPU), a Field Programmable Gate Array (FPGA), another processor, or any suitable combination thereof. The term "processor" may include multi-core processors that may comprise two or more independent processors (sometimes referred to as "cores") that may execute instructions contemporaneously. These cores can be homogeneous (e.g., all cores are identical, as in multicore CPUs) or heterogeneous (e.g., cores are not identical, as in many modern GPUs and some CPUs). In addition, the term "processor" may also encompass systems with a distributed architecture, where multiple processors are interconnected to perform tasks in a coordinated manner. This includes cluster computing, grid computing, and cloud computing infrastructures. Furthermore, the processor may be embedded in a device to control specific functions of that device, such as in an embedded system, or it may be part of a larger system, such as a server in a data center. The processor may also be virtualized in a software-defined infrastructure, where the processor's functions are emulated in software.

"Signal medium" may include, for example, an intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term "signal medium" shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term "modulated data signal" means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms "transmission medium" and "signal medium" mean the same thing and may be used interchangeably in this disclosure.

"User device" may include, for example, a device accessed, controlled or owned by a user and with which the user interacts to perform an action, engagement or interaction on the user device, including an interaction with other users or computer systems.

<u>What is claimed is:</u>

1. A system comprising:

at least one processor; and

at least one memory component storing instructions that, when executed by the at least one processor, cause the at least one processor to perform operations comprising:

retrieving, from a third-party media streaming service via an application programming interface, media consumption activity data associated with a first user of a social interaction platform;
converting the media consumption activity data into a standardized activity record according to a unified schema;
determining, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service; and
causing display, within a surface of the social interaction platform presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record.
2. The system of claim 1, wherein the operations further comprise:
executing an OAuth-based authorization flow with the third-party media streaming service on behalf of the first user, the OAuth-based authorization flow comprising transmitting an authorization request to the third-party media streaming service and receiving an access token in response to the first user granting consent;
storing the access token in association with an account of the first user on the social interaction platform;
detecting that the access token has expired or been invalidated by the third-party media streaming service; and
executing a token refresh operation to obtain a replacement access token, or, in response to determining that the token refresh operation has failed, transmitting a notification to the first user indicating that a connection to the third-party media streaming service is no longer active.
3. The system of claim 1, wherein the surface comprises a map interface, and wherein causing display of the listening status indicator comprises:
determining a geographic location of the first user based on one or more location signals received from a device of the first user;
rendering an avatar representing the first user at a position on the map interface corresponding to the geographic location of the first user;
generating a callout element positioned proximate to the avatar, the callout element comprising a track name and an artist name extracted from the standardized activity record; and
rendering, within the callout element, a selectable element that, upon selection by the second user, initiates navigation to a content page associated with a track identified in the standardized activity record.
4. The system of claim 3, wherein the operations further comprise:
detecting a selection by the second user of the avatar or the callout element on the map interface;
in response to the detecting the selection, expanding a tray element within the map interface, the tray element comprising a profile identifier of the first user, a last-active timestamp associated with the first user, and the listening status indicator;
rendering, within the tray element, one or more interactive elements comprising at least an add element configured to initiate a save operation for a track identified in the standardized activity record; and
rendering, within the tray element, a streaming service identifier indicating the third-party media streaming service from which the media consumption activity data was retrieved.
5. The system of claim 4, wherein the operations further comprise:
detecting a selection of the add element by the second user;
identifying, from the standardized activity record, a track identifier associated with the track;
transmitting, via a second application programming interface associated with a streaming service linked to an account of the second user, a write request to save the track to a library of the second user on the streaming service linked to the account of the second user; and
in response to receiving a confirmation that the write request was processed, updating the add element from a first state indicating an available save action to a second state indicating that the track has been saved, and causing display of a confirmation notification on the device of the second user.
6. The system of claim 5, wherein the streaming service linked to the account of the second user is different from the third-party media streaming service associated with the first user, and wherein the operations further comprise:
extracting an international standard recording code from the standardized activity record associated with the track;
matching the international standard recording code to a corresponding track entry in an internal media library of the social interaction platform;
resolving, from the corresponding track entry, a platform-specific track identifier for the streaming service linked to the account of the second user; and
transmitting the write request to the streaming service linked to the account of the second user using the platform-specific track identifier resolved from the corresponding track entry.
7. The system of claim 3, wherein the operations further comprise:
determining whether a viewing user to whom the map interface is being presented corresponds to the first user or to a user other than the first user;
in response to determining that the viewing user corresponds to the first user, rendering, within the callout element or the listening status indicator associated with the first user on the map interface, a platform-specific icon identifying the third-party media streaming service from which the media consumption activity data was retrieved; and
in response to determining that the viewing user corresponds to a user other than the first user, rendering, within the callout element or the listening status indicator associated with the first user on the map interface, a generic media icon in place of the platform-specific icon, the generic media icon being visually distinct from the platform-specific icon and not identifying a particular third-party media streaming service.
8. The system of claim 1, wherein the operations further comprise:
retrieving, from a second third-party media streaming service different from the third-party media streaming service, second media consumption activity data associated with a third user of the social interaction platform;
parsing the second media consumption activity data using a partner-specific interface module associated with the second third-party media streaming service, the partner-specific interface module mapping one or more fields of the second media consumption activity data to corresponding fields of the unified schema; and
converting the second media consumption activity data into a second standardized activity record according to the unified schema, the second standardized activity record comprising a partner identifier distinguishing the second third-party media streaming service from the third-party media streaming service, wherein the determining the presence state and the causing display are performed using the unified schema regardless of which third-party media streaming service provided the media consumption activity data.
9. The system of claim 8, wherein the unified schema comprises one or more fields selected from a group consisting of a user identifier, a track identifier, an artist identifier, an album identifier, a playback state, a timestamp, a session identifier, and a partner identifier, and wherein converting the media consumption activity data into the standardized activity record comprises:
parsing a response received from the application programming interface of the third-party media streaming service to extract one or more partner-specific data fields;
mapping each of the one or more partner-specific data fields to a corresponding field of the unified schema; and
populating the standardized activity record with values derived from the mapping.
10. The system of claim 1, wherein determining the presence state comprises:

comparing the timestamp of the standardized activity record against a current time to compute an elapsed duration;
in response to determining that the elapsed duration is below a first threshold and the playback state indicates active playback, classifying the presence state as a live listening state;
in response to determining that the elapsed duration is above the first threshold and below a second threshold, classifying the presence state as a recently listened state; and
in response to determining that the elapsed duration exceeds the second threshold, classifying the presence state as an inactive state and suppressing display of the listening status indicator for the first user.
11. The system of claim 10, wherein the operations further comprise:
monitoring an application activity signal associated with the first user on the social interaction platform, the application activity signal indicating a last time the first user opened or interacted with the social interaction platform;
determining that a duration since the last time the first user opened or interacted with the social interaction platform exceeds an inactivity threshold;
in response to determining that the duration exceeds the inactivity threshold, ceasing retrieval of the media consumption activity data from the third-party media streaming service for the first user; and
in response to detecting a subsequent opening of the social interaction platform by the first user, resuming retrieval of the media consumption activity data from the third-party media streaming service and restoring a prior sharing configuration associated with the first user.
12. The system of claim 11, wherein the operations further comprise:
detecting that the first user has reopened the social interaction platform after the duration exceeded the inactivity threshold;
determining that the first user is actively consuming media on the third-party media streaming service at a time of the reopening;
generating a reminder notification for display within the social interaction platform on the device of the first user, the reminder notification informing the first user that the listening status indicator is visible to one or more friends of the first user; and
presenting, within the reminder notification, one or more selectable options comprising at least an option to continue sharing and an option to activate a private listening mode.
13. The system of claim 1, wherein the operations further comprise:
receiving, from the first user, a sharing preference selection, the sharing preference selection specifying one or more visibility rules for the listening status indicator, the one or more visibility rules comprising at least one rule selected from a group consisting of sharing with all friends of the first user, sharing with a subset of friends with whom the first user shares location data, and sharing with a custom list of one or more individually selected friends;
storing the sharing preference selection in association with an account of the first user;
prior to causing display of the listening status indicator, evaluating whether the second user satisfies the one or more visibility rules specified by the sharing preference selection; and
causing display of the listening status indicator to the second user only in response to determining that the second user satisfies the one or more visibility rules.
14. The system of claim 13, wherein the operations further comprise:
receiving, from the first user, a selection of a private listening mode, the selection specifying a duration parameter, the duration parameter indicating one of a fixed time period or an indefinite period;
in response to receiving the selection of the private listening mode, suppressing display of the listening status indicator for the first user to all friends of the first user regardless of the sharing preference selection;
ceasing retrieval of the media consumption activity data from the third-party media streaming service for the first user for the duration specified by the duration parameter; and
in response to determining that the duration parameter has elapsed, deactivating the private listening mode and resuming retrieval of the media consumption activity data and display of the listening status indicator according to the sharing preference selection.
15. The system of claim 1, wherein causing display of the listening status indicator comprises rendering the listening status indicator on one or more surfaces of the social interaction platform, and wherein the operations further comprise at least one of:
rendering the listening status indicator within a map interface as a callout element positioned proximate to an avatar of the first user;
rendering the listening status indicator within a profile interface of the first user as a persistent indicator element displaying at least a track name and an artist name from the standardized activity record;
rendering the listening status indicator within a contextual tray element that is displayed in response to a selection interaction by the second user on the map interface; or
rendering the listening status indicator within a content page associated with a media track identified in the standardized activity record.
16. The system of claim 1, wherein the operations further comprise:
detecting that the second user has not linked a streaming service account to the social interaction platform;
detecting an interaction by the second user with the listening status indicator associated with the first user, the interaction comprising a selection of a save element rendered within the listening status indicator;
in response to detecting the interaction, presenting a connection prompt to the second user, the connection prompt comprising one or more selectable options each corresponding to a different third-party media streaming service available for linking;
initiating an authorization flow with a third-party media streaming service selected by the second user from the one or more selectable options; and
logging the interaction and a surface from which the connection prompt was presented as a conversion event.
17. The system of claim 16, wherein the operations further comprise:
presenting the connection prompt from one or more entry points within the social interaction platform, the one or more entry points comprising at least two of a map tray element, a content page associated with a media track, a settings interface, a profile indicator element, or a feed banner element;
recording, for each presentation of the connection prompt, an entry point identifier indicating which of the one or more entry points triggered the presentation;
computing a conversion rate for each entry point identifier based on a ratio of completed streaming service linkings to presentations of the connection prompt from that entry point; and
adjusting a frequency or a priority of presenting the connection prompt at each of the one or more entry points based on the computed conversion rates.
18. The system of claim 1, wherein the operations further comprise:
detecting a selection by the second user of the listening status indicator or a portion thereof;
generating a deep link comprising one or more parameters identifying a track indicated in the standardized activity record, the one or more parameters comprising at least a partner-specific track identifier compatible with the third-party media streaming service or a streaming service linked to the

second user;
determining whether a native application associated with a target streaming service is installed on the device of the second user;
in response to determining that the native application is installed, transmitting the deep link to the native application to cause the native application to navigate to a content view associated with the track; and
in response to determining that the native application is not installed, redirecting the second user to a web-based interface or an application store listing associated with the target streaming service.
19. A computer-implemented method comprising:
retrieving, from a third-party media streaming service via an application programming interface, media consumption activity data associated with a first user of a social interaction platform;
converting the media consumption activity data into a standardized activity record according to a unified schema;
determining, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service; and
causing display, within a surface of the social interaction platform presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record.
20. A non-transitory computer-readable storage medium storing instructions that, when executed by at least one processor, cause the at least one processor to perform operations comprising:
retrieving, from a third-party media streaming service via an application programming interface, media consumption activity data associated with a first user of a social interaction platform;
converting the media consumption activity data into a standardized activity record according to a unified schema;
determining, based at least on a playback state and a timestamp of the standardized activity record, a presence state for the first user, the presence state indicating a status of the first user's media consumption on the third-party media streaming service; and
causing display, within a surface of the social interaction platform presented on a device of a second user, of a listening status indicator associated with the first user, the listening status indicator comprising at least a portion of the standardized activity record.

Abstract

A system retrieves media consumption activity data from a third-party media streaming service for a first user of a social interaction platform, converts the data into a standardized format, determines the user's presence state based on the playback state and timestamp, and displays a listening status indicator on a device of a second user, showing the first user's media consumption status.